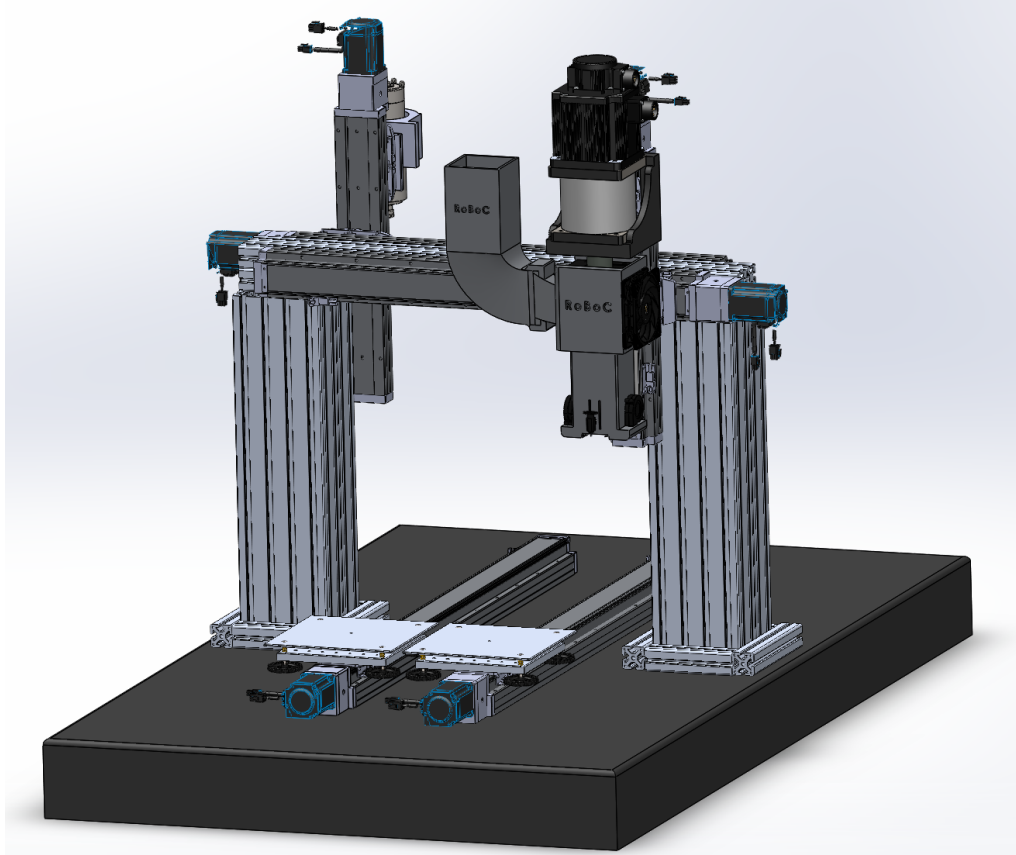




Development of a Hybrid Manufacturing System for High Viscosity Ceramic Composites: Integrating Robocasting and CNC Machining



BEng PROJECT FINAL DISSERTATION 2023-24

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I extend my sincere gratitude to all those who supported the development of this dissertation. The work presented in this report has been conducted solely by me, [REDACTED], except for the subtractive manufacturing section. This includes the CNC X, Y, Z axes, spindle, design, control, and wiring. The rest of the work, specifically related to the additive manufacturing and synchronization of the additive manufacturing part with the subtractive manufacturing part of the hybrid manufacturing machine, has been accomplished solely by myself.

Summary

The development of the Hybrid Manufacturing (HM) machine is a pivotal advance in the ceramic additive manufacturing sector, aimed at enhancing the production of high-viscosity advanced ceramic composites, essential in aerospace, biomedical, and electronics industries. The project's goal was to design a system that integrates robocasting with CNC machining layer-by-layer to overcome the limitations of traditional manufacturing methods by improving precision and surface quality.

The objectives were to engineer a robust extrusion section capable of handling viscosities up to 40,000 pa.s at 0.3 s⁻¹, develop precise XYZ motion control, establish seamless synchronization between additive and subtractive manufacturing processes, and validate the system through rigorous testing. These goals were meticulously addressed through extensive design, assembly, and testing phases.

Key experiments included the Position Accuracy XY Experiment, which confirmed the X-Y axes' precision within ± 5 micron. The Temperature Stability Experiment demonstrated the system's ability to maintain temperature fluctuations within $\pm 2^\circ\text{C}$ across critical heater zones, ensuring consistent material properties during processing. The Extrusion Capability Experiment verified the extruder's effectiveness, consistently producing filament with an average thickness deviation of only 10 microns with die cooling mechanisms engaged and forming a 10x10x10 mm cube with dimensional accuracies of 585 microns for the X-axis, 535 microns for the Y-axis and 20 microns for the Z-axis of the printed cube.

The HM machine's novel synchronization capability was showcased in a test that efficiently managed simultaneous operations of printing and CNC machining, illustrating a substantial reduction in manufacturing times and enhanced operational efficiency. The synchronization, facilitated by custom-developed software, aligned the additive and subtractive processes, enabling the machine to perform complex tasks with minimal manual intervention.

This study's significant contribution lies in its successful demonstration of a hybrid manufacturing approach that integrates layer-by-layer CNC machining and robocasting. This integration not only meets but advances the current manufacturing standards by providing a means to produce complex ceramic components with high accuracy and superior surface finishes. The HM machine's capabilities, such as handling material viscosity up to 40,000 pa.s and maintaining precise control over extensive manufacturing operations, underline its potential to reshape the future of advanced ceramics manufacturing.

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1. INTRODUCTION

1.1. Background and Context

Additive manufacturing (AM), commonly known as 3D printing, involves fabrication methods that add material layer-by-layer to create objects with complex geometries. These methods can be categorized as either direct or indirect, based on the material deposition technique [1].

Ceramic materials are utilized in industries such as aerospace[2], electronics, machinery, and biomedical [3] applications. Their adoption is due to unique mechanical and functional properties like high strength, thermal stability, conductivity, and magnetic performance [4]. Despite the economic challenges posed by the pandemic, the ceramic additive manufacturing industry has shown remarkable resilience and growth. In 2021, it achieved a global revenue of \$402 million and is projected to surge to \$3.678 billion by the year 2028 [5, 6].

Given the increasing structural complexity of ceramic components, additive manufacturing has emerged as a viable method for producing these parts in a single run, offering advantages over traditional manufacturing techniques.

Among additive manufacturing techniques, Stereolithography (SL) and Digital Light Processing (DLP) are typically employed for indirect deposition of ceramic slurries. Despite their fine feature resolution, these techniques face limitations due to the rheological properties of the highly loaded ceramic slurry, typically <41 %vol [7], and the refractive index of the ceramic particles. Higher particulate loading restricts the range of printable viscosity, subsequently enhancing the green density and influencing the functional and mechanical attributes of the sintered final product [8]. Additionally, many advanced ceramics are incompatible with SL and DLP methods due to their high refractive index, which renders them unresponsive to photopolymerization and curing by these light-based techniques.

Direct Ink Writing (DIW), also known as "robocasting," employs extrusion techniques that utilize highly concentrated powder suspensions, or inks [1]. Unlike photopolymerization-based methods, robocasting accommodates high particulate loading and a wide range of viscosities, depending on the extruder design. This flexibility enables the printing of a diverse array of ceramic composites as well as metals. As a result, the printed components exhibit higher density, which substantially enhances their functional and mechanical properties [9]. The main disadvantage of robocasting is that parts will always have step edges due to the layer wise nature of manufacture. The size of the step edges are defined by the radius of the nozzle (or die), so are typically above 100 μm , while other techniques such as stereolithography can achieve a fifth of that size [9].

To overcome the limitations associated with robocasting, particularly those concerning surface finish, the incorporation of subtractive manufacturing techniques, such as Computer Numerical Control (CNC), can be advantageous.

In the realm of Subtractive Manufacturing, CNC technology utilizes microcomputers to create, interpret, and carry out the sequential commands that dictate the actions of the end effector [10]. Integrating CNC technology mitigates the surface finish issues inherent to robocasting.

Hybrid Manufacturing (HM) has evolved from its initial definition by Rajurkar et al. [11], which emphasized automation, to a more inclusive, digitally-driven framework that integrates multiple types of manufacturing processes[12, 13]. While early research by Greul et al. [14] focused on metallic parts, current studies have expanded to include diverse materials like gypsum-based ceramics [15]. However, a gap remains in addressing high-viscosity ceramic composites. The present study aims to fill this void by developing a hybrid machine tailored for these specific materials, thereby contributing to the growing versatility of HM technologies to improve efficiency and accuracy for creating complex, customized items.

The aim of this work is to develop a Hybrid Manufacturing machine, which will combine robocasting and CNC machining to create thermoplastic-based, high-load ceramic composites. Unlike existing commercial options, this HM machine offers layer-by-layer CNC finishing allowing for CNC machining of intricate areas that are otherwise inaccessible. Previous studies by researchers like Mertkan [16] and Lee [17], have explored hybrid manufacturing systems optimized for thermoplastics, notably PLA. In contrast, the aim of the current research is to develop a machine designed specifically for high-viscosity ceramic composites. Although this additional step extends the production timeline, the machine is engineered to perform robocasting and CNC machining simultaneously, thereby optimizing the process for industrial scalability.

1.2. Aim and Objectives

The aim of this work is to develop a Hybrid Manufacturing machine that combines robocasting and CNC machining to fabricate high-load ceramic composites. The goal is to address limitations in existing manufacturing methods and improve material properties. This report will mainly focus on additive manufacturing aspect, as well as synchronization between the CNC and extrusion sections.

To achieve this aim, the following specific objectives have been outlined:

- **Extrusion Section Design:** Complete the engineering and assembly of the extrusion section, capable of handling high-viscosity ceramic composites.
- **Extruder Motion Control Design:** Develop and test an XYZ motion control for the additive manufacturing part of the machine with deviation less than 50 micron.
- **CNC-Extruder Synchronization:** Develop a protocol for the X-axis control transfer between the additive and subtractive sections of the hybrid manufacturing machine to ensure synchronized operation. Validate the mechanism to confirm smooth transition and control handover between the additive and subtractive sections.
- **Testing and Validation:** Conduct essential tests to measure the extruder system's reliability and the effectiveness of the X-axis control transfer synchronization.

2. LITERATURE REVIEW

In the evolving landscape of manufacturing technologies, hybrid additive manufacturing techniques have emerged as a significant area of progress and innovation. Initiated by Greul et al., the field has seen substantial advancements through the contributions of Grguraš and Kramar [18] and Li et al. [19], who integrated Fused Deposition Modelling (FDM) with milling and CNC technology, significantly enhancing efficiency and reducing production times. Jeongsu Lee [17] further expanded this with a high-speed, large-scale system for mold fabrication. Additionally, J. Hinton's research [6] on integrating ceramic composites using robocasting with CNC machining has underscored the potential for precision in manufacturing ceramic composites. This study aims to build on these foundations, addressing the precise, layer-by-layer CNC machining of ceramic composites with a novel and efficient 3-axis movement strategy [20].

2.1. Feeding Material Type

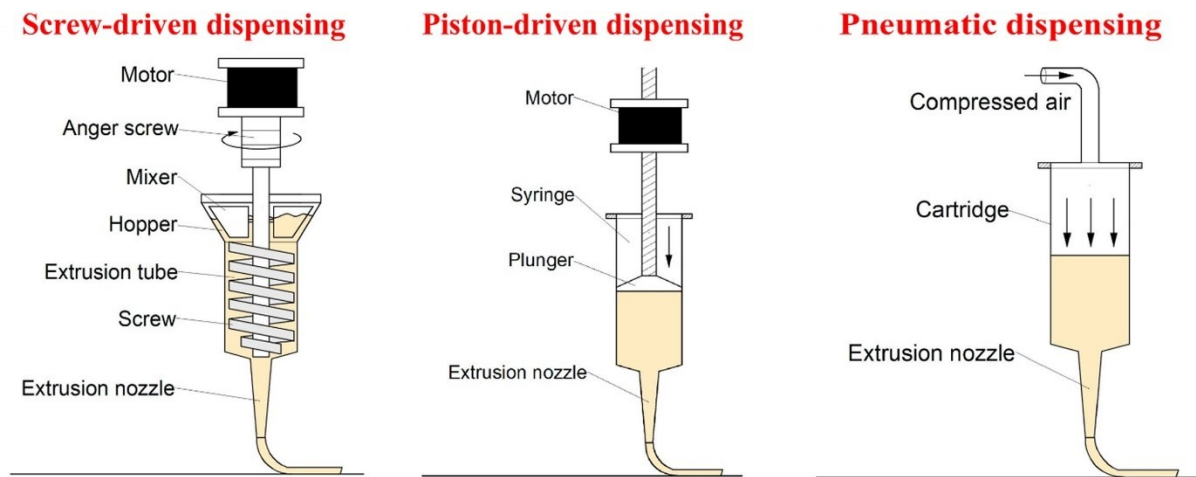
In the realm of robocasting (DIW), the composition of ceramic composites is quite diverse, typically comprising ceramic powder, various additives, and a bulk material. The bulk material plays a pivotal role, often being a solvent that imparts a paste-like phase to the composite, or a polymer binder [21-23]. The choice of binder is critical, as it determines the composite's state – solid, called pellet, or liquid paste – contingent upon the polymer's glass transition and melting temperature. Many researchers, including Peter Veteška [21], have advocated the use of ceramic composite pellet feedstock for extrusion additive manufacturing. Pellets additive manufacturing (PAM) utilization is noted for its lower environmental impact, reduced energy consumption, and simplified material procurement [23, 25], making it more suitable for the highly scalable application of this device in industry. Similarly, Samuel E. Hall [22] and others have explored paste feedstock for its capability to print non-organic (non-polymer) materials. This study primarily investigates polymer feeding using pellets, focusing on the advantages of their cost-effectiveness and ease of access, while also considering the environmental and energy-efficient benefits of this approach.

2.2. Extruder Design

2.2.1. Extrusion Method

In robocasting's Direct Ink Writing (DIW), selecting the right extrusion system is essential. Figure 1 shows the options: piston-driven, pneumatic, and screw-driven systems. S. Lamnini notes [1] the simplicity of the first two setups, while J.-W. Tseng [23] highlights the screw-driven system's advantage in minimizing backflow and air entrapment, crucial for precise control in processing highly loaded powder suspensions. Chao-Fan Guo's research [24] shows material properties greatly affect screw-driven printing outcomes, while Medeni Maskan [26] notes such systems struggle with high-slip materials due to factors like surface energy and viscosity. These findings emphasize the critical need to select an extrusion system that aligns with the specific characteristics of the material to optimize the printing process.

This study employs a screw-driven system for its efficient, precise handling of ceramic composites, validated by Qinglong He's findings [25] on enhancing product quality in PAM and robocasting.



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Figure 1: Extrusion setups for viscous pastes, showing screw-driven, piston, and pneumatic systems for ink flow.

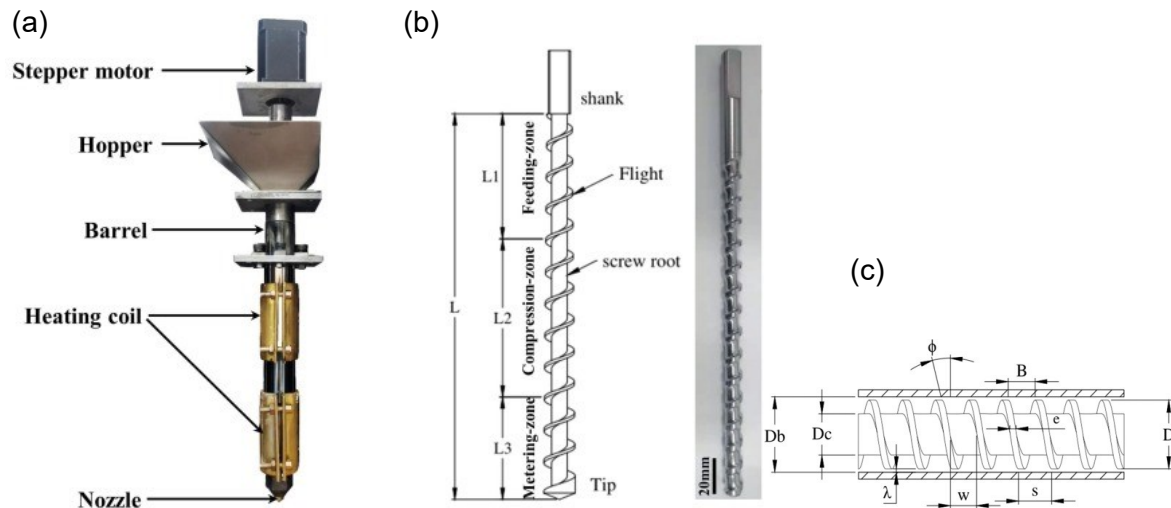
2.2.2. Screw Type and Dimension

Alessio Pricci's research [24] suggests that combining insights from Injection Molding, Food [26], and Plastic Extrusion [29] helps identify the right screw type and dimensions for DIW applications.

In robocasting, as depicted in Figures 1 and 2, the choice between single-screw and twin-screw extruders is crucial. Giles Jr. [27] notes that single-screw extruders are simpler and more cost-effective, while twin-screws excel in mixing efficiency.

Following the choice of single-screw and twin-screw extruders, a key focus is their dimensions, illustrated in Figure 2(c) [26]. Screws are defined by diameter (D) and length (L). M. Maskan [26] and F. Giles Jr. [27] highlight the aspect ratio's (L/D) impact on performance: a lower ratio reduces torque and power needs, while a higher ratio improves mixing and melting for materials like ceramic composites. The screw compression ratio ($\epsilon = h_F / h_M$) also plays a vital role in processing polymeric materials.

In line with Qinglong He's suggestions [25], the aspect ratio for this research is chosen within the range of 16:1 to 24:1 and the compression ratio between 2:1 to 3:1. This configuration is specifically tailored for ceramic polymer-based composites, ensuring adequate melting time for the ceramic pellet feedstock and effective elimination of entrapped air, thus achieving a stable flow through the nozzle.



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Figure 1. Feedstock Delivery System in FDM Machine: (a) showcases the extrusion device, while (b) details the structure of the screw [25] (c) details the geometry variables of a screw[1].

2.2.3. Single Screw Investigation

M. Maskan [26] and Q. He [25] describe a typical single screw extruder, shown in Figure 2, as having three sections: feeding, compression, and metering zones, for pellet conveyance, melting, and molten material transport, respectively. Their lengths are usually 4-8, 4-5, and 6-10 times the screw's diameter.

S. Levy [31] points out the low thermal conductivity challenge in polymer composites, extending melting times. Q. He's research [25], however, demonstrates that incorporating ceramics with higher thermal conductivity can overcome this, underscoring the suitability of single screw extrusion for ceramic composites. This leads to a more efficient screw design, as visually explained in Figure 3, which illustrates the melting process within the single screw.

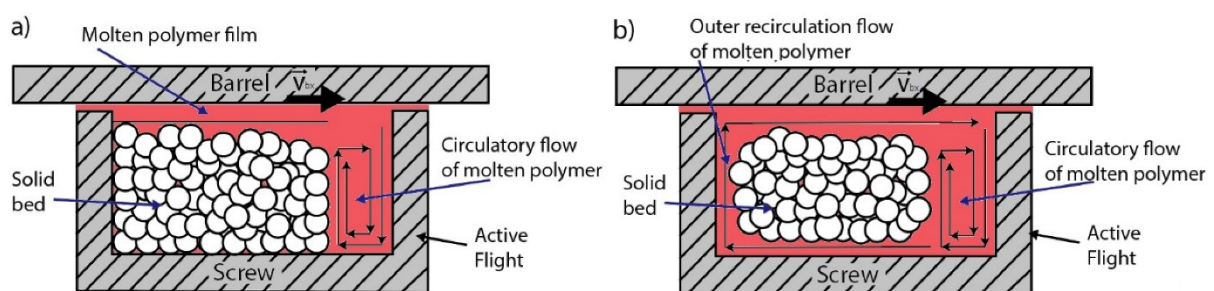


Figure 2. Melt Removal via Screw Mechanism: (a) as documented by Maddock and Tadmor [26] for Single Screw Extruders; (b) depicts the outer recirculation flow noted in multiple instances within the screw micro-extruder [27].

Giles Jr. [28] and S. Levy [29] note in their works that in robocasting, the metering zone's role is crucial for transporting polymer melt and determining melt flow rate. Influenced by material properties and screw design, including drag (Q_d) and pressure (Q_p) flow rates, this zone directly impacts the extrusion outcome, as shown in Figure 4.

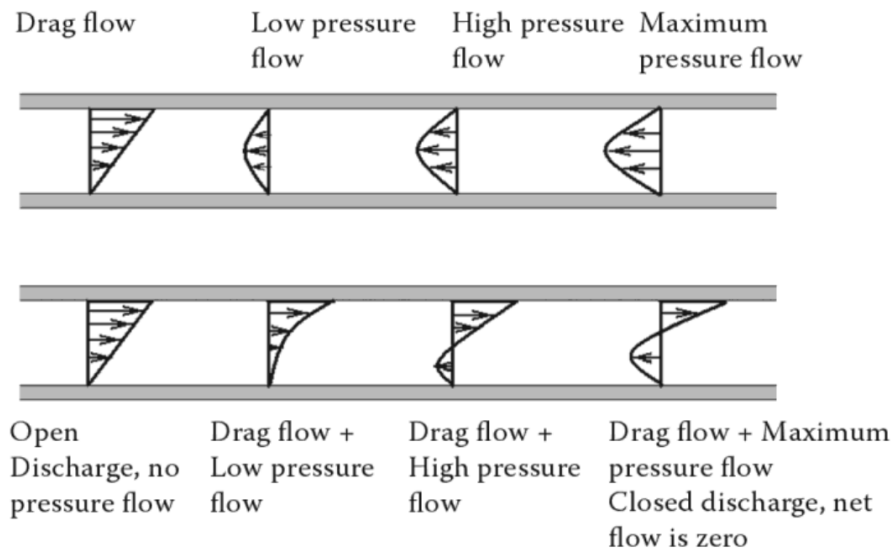


Figure 3. Illustration of Drag Flow, Pressure Flow, and Combined Flow in the Context of Escalating Pressure Conditions [30].

2.2.4. Extruder Control

The realm of control algorithms in single screw extrusion for robocasting is marked by a diverse array of methodologies, each tailored to address specific aspects of the process. Temperature and pressure control are fundamental for ensuring process stability and quality of the output. Studies by Dörmeier [31] and Fingerle [32] employing PID controllers have demonstrated effectiveness in managing long-term disturbances, a common challenge in extrusion processes. However, their efficacy can vary with the nature and frequency of these disturbances. On the other hand, Muhrer et al. [33] employ a Cascade PI approach, offering a nuanced control mechanism that caters to varying operational conditions. This adaptability is crucial in processes where consistent quality needs to be maintained despite fluctuations in external or internal process variables.

In the context of viscosity control, a key factor in determining the final product's consistency and quality, approaches like the Fuzzy-PID by Chiu et al. [34] stand out. This method, particularly when combined with in-line viscometry, offers a more responsive and precise control, crucial for materials with complex rheological properties. Additionally, Guo et al. [35] introduce a combination of PI with if-then rules and pattern recognition, signalling a move towards more intelligent, adaptive control systems. Such systems are designed to maintain dimensional stability of products, a vital aspect in high-precision manufacturing. The diversity and specificity of these control strategies emphasize the necessity of a tailored approach in selecting control systems, one that aligns closely with the unique requirements of the extrusion process, material properties, and desired product attributes.

The literature review highlights significant advancements in extrusion methods and hybrid additive manufacturing, notably in the integration of extrusion with CNC machining, as demonstrated by Grguraš and Kramar [18], Li et al. [19], and Jeongsu Lee [17]. These

methods generally involve printing the entire structure before CNC machining the periphery, a process that limits precision in machining intricate, layer-by-layer areas post-construction. This highlights a critical research gap for developing precise, layer-by-layer CNC machining capabilities, especially for ceramic materials. Addressing this, the proposed study introduces a novel 3-axis movement strategy [20] that enhances CNC machining's precision and accessibility. Leveraging a single screw extruder, optimized through literature insights for screw geometry and flow rate control, this approach not only ensures optimal print quality but also elevates the quality of final ceramic components in this hybrid manufacturing setup.

3. METHODOLOGY

The Hybrid Manufacturing (HM) machine integrates the flexibility of robocasting with the precision of CNC machining, creating a robust platform for producing dense ceramic components. Illustrated in the CAD model (Figure 4), it features a sturdy aluminium frame that accommodates both the additive and subtractive processes on a granite base to ensure stability and precision.

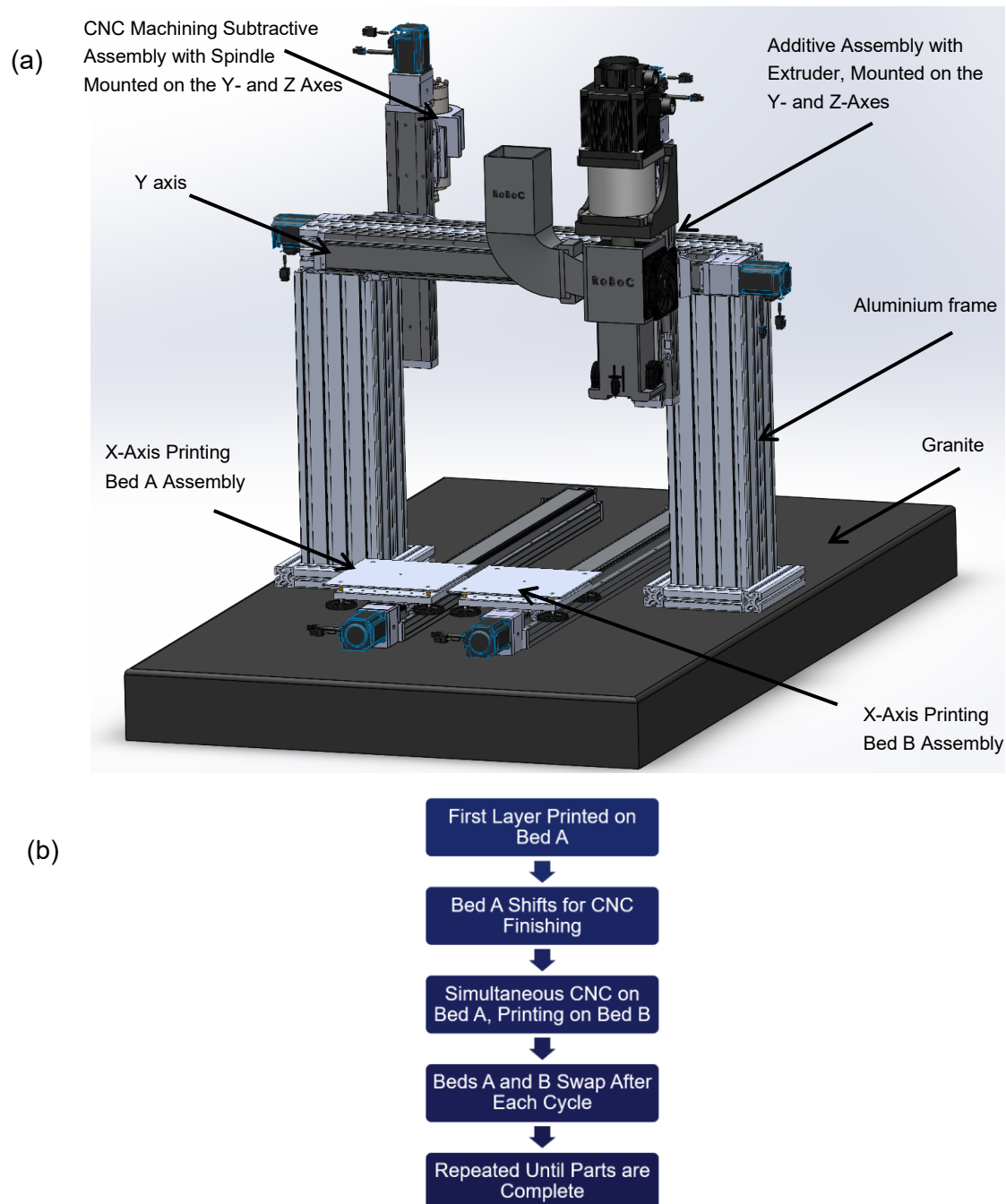


Figure 4 – (a) CAD Model of the Hybrid Manufacturing Machine, detailing its core components and assembly structure. (b) Operational Flowchart, illustrating the synchronized additive manufacturing and CNC machining process cycle.

Operationally, the machine leverages a synchronized methodology—sequentially printing on one bed while conducting CNC machining on another. This strategy allows for seamless production flow and the efficient crafting of complex geometries made of advanced ceramic materials, which are challenging for conventional manufacturing methods.

Central to its innovation, the HM machine boasts an extruder system specifically designed to process high-viscosity (up to 4000 pa.s at 0.3 s^{-1} shear rate) ceramic-binder materials. This capability expands the range of materials that can be used, resulting in denser products with enhanced mechanical and functional properties. Additionally, the integration of CNC machining not only refines the surface finish of the end products but also significantly improves their dimensional accuracy.

This harmonized approach, bridging advanced material handling with precision finishing, ensures the HM machine not only meets its set objectives but also propels the field of ceramic manufacturing to new heights. This overview sets the stage for what follows: a dive into initial proof of concept, detailed component design, and the machine's control systems, all ensuring the HM machine's precision and efficiency.

3.1. Initial Tests and Proof of Concept

The printability of material supplied by AM-COE, along with its machinability, was gauged using initial extrusion and machining experiments. As shown in Figure 6, a syringe extruded the material through nozzles of 1mm and 0.4mm diameters. Trials with the 0.4mm nozzle were discontinued due to the high pressures required for extrusion, underscoring the necessity for a single screw and barrel system capable of generating the high pressure needed to extrude through smaller diameters. Conversely, the 1mm nozzle was effectively used, facilitated by comparatively lower pressure demands. The temperature-induced shrinkage, visible in Figure 6, caused some extruded material to break or detach from the print bed, emphasizing the need for a heated bed and a controlled temperature environment for printing.

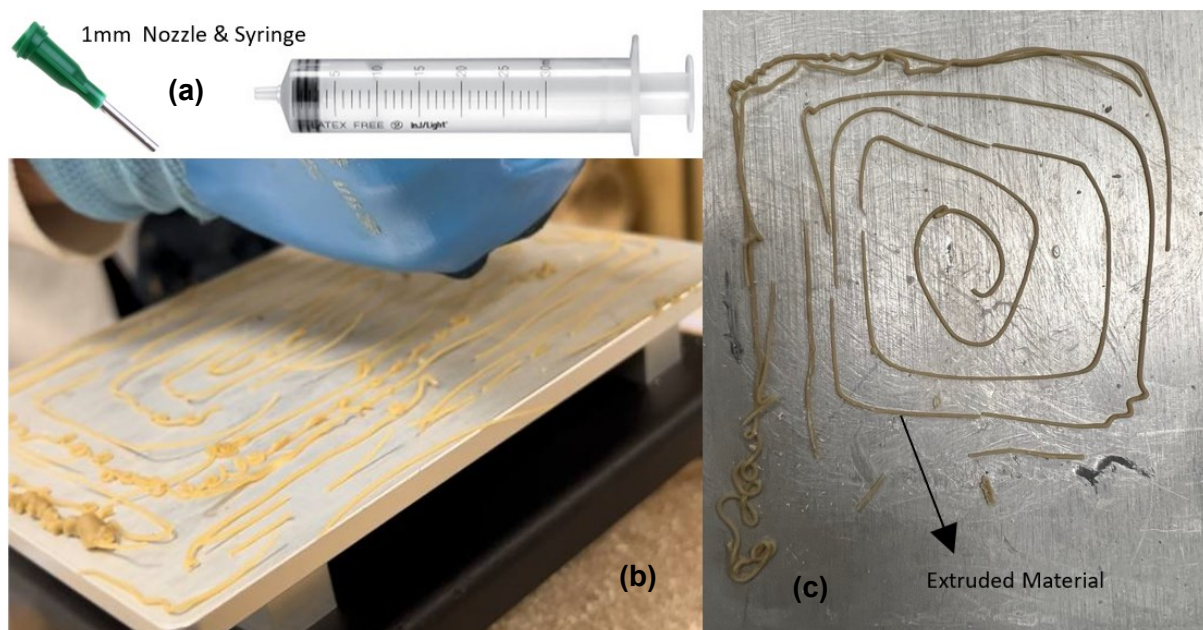


Figure 5. Material Extrusion Experiment: (a) shows the syringe used in this experiment with the 1mm nozzle; (b) depicts the extrusion in progress, and (c) highlights the material after deposition. The test validates the effective operation of the 1mm nozzle at manageable pressures and rules out the 0.4mm nozzle due to excessive pressure requirements.

For the CNC machining aspect, a straightforward setup was utilized, featuring a drill equipped with a polishing tool to machine the outer surface of the extruded layer. The machinability test demonstrated the viability of performing layer-by-layer machining immediately after extrusion. Additionally, it was noted that controlling the printer room temperature is essential to adjust the material layer's stiffness for optimal machinability.

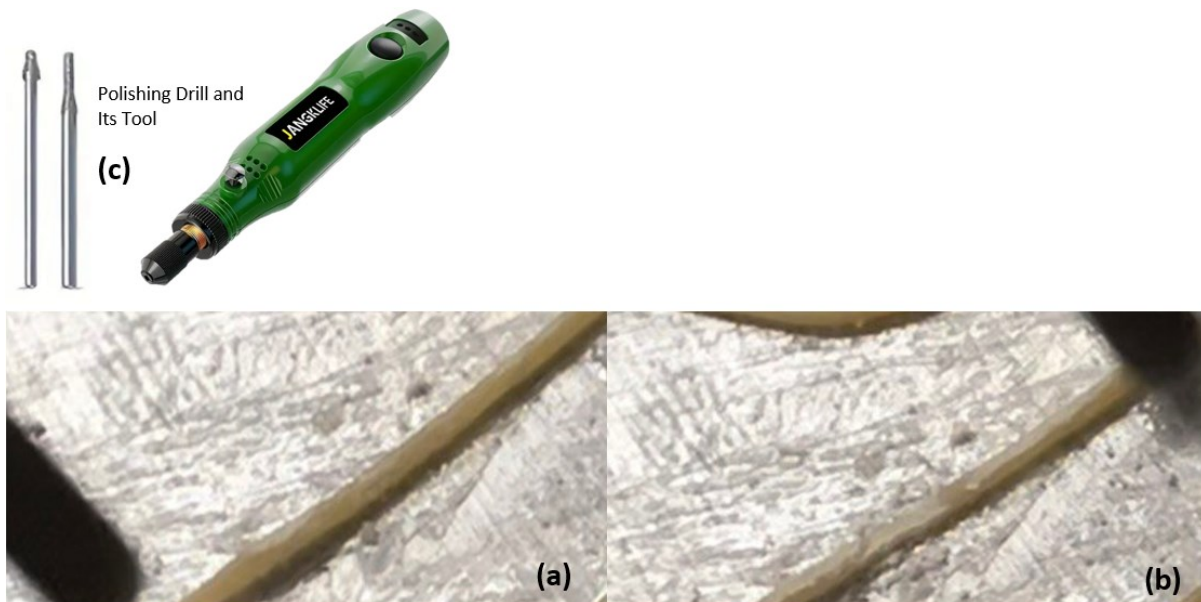


Figure 6. CNC Machining Test Overview: (a) presents the surface texture before machining; (b) shows the surface post-machining; and (c) features the polishing drill used. This sequence highlights the viability of in-situ, layer-by-layer machining in additive manufacturing, underscoring the crucial role of temperature control for optimal results.

3.2. Extruder Assembly Design

The extruder assembly, a crucial component of the Hybrid Manufacturing machine, is meticulously engineered to process high-viscosity ceramic composites. As shown in Figure 7, beginning with the hopper, ceramic pellets are funnelled into the feeding section of the screw extruder, featuring a 12 mm diameter screw designed to compress and convey the material towards the nozzle. The screw geometry is precision-crafted to apply pressure and induce melting, transforming the pellets into a malleable paste.

En route to the nozzle, the paste is heated by three strategically placed heat bands, capable of reaching up to 400 degrees Celsius. This heat is essential for melting the polymer matrix, enhancing the flowability of the ceramic-polymer paste through the nozzle. Concurrently, a heat sink—visible in Figure 7(c)—acts as a thermal barrier. Coupled with a feeding fan, it dissipates heat from the feeding section, preventing heat transfer to the gearbox and motor. This cooling mechanism is vital for maintaining the integrity of the feed and preventing clogging caused by the bridging of pellets.

To address post-extrusion challenges, two die fans are situated at the nozzle's exit, rapidly cooling the extruded material to mitigate die swell effects, ensuring dimensional stability of the printed layer.

The integration of a planetary gearbox serves to amplify the motor's torque, ensuring consistent extrusion under high load conditions. Lastly, a levelling probe is employed to finely adjust the extruder's position relative to the printing bed, optimizing the accuracy and quality of the extruded layers.

This assembly reflects a harmonious balance between heating elements and cooling systems, all orchestrated to maintain a steady state of extrusion conducive to high-quality manufacturing.

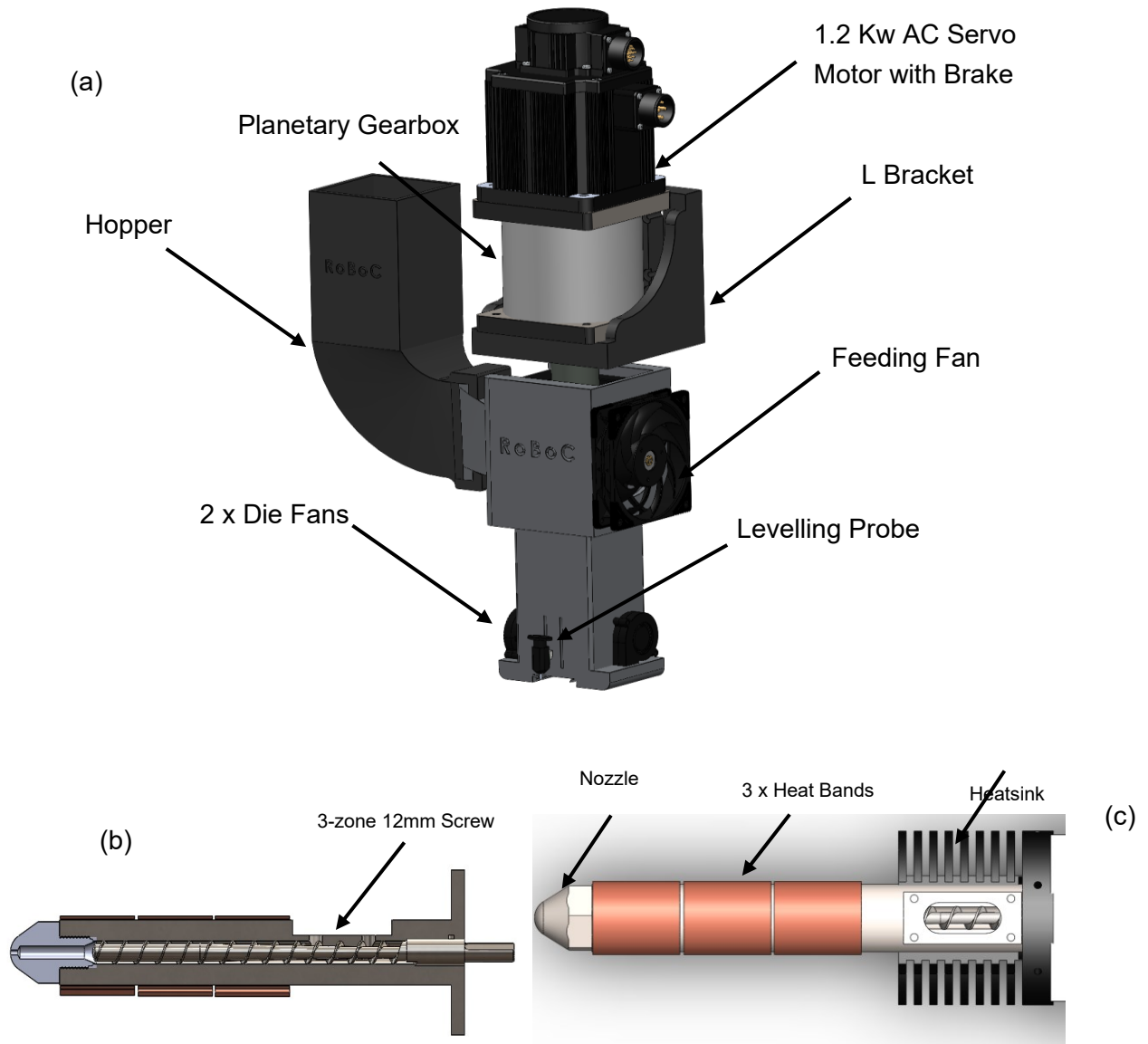


Figure 7- (a) Complete extruder assembly with key components labelled. (b) Cross-sectional view of the 12mm diameter screw. (c) Nozzle with heat bands and cooling heatsink.

3.2.1.Screw Design Calculations

The design of the screw is crucial for the extruder in a Hybrid Manufacturing (HM) system, a conclusion drawn from the literature review.

In the simplified extrusion process model, several key assumptions were made: the process is in steady state with constant polymer density and thermal conductivity, the polymer melt flow is laminar with minimal effects from inertia and body forces, wall slip at the flow channel interface is absent, the melt's viscosity is Newtonian and unaffected by temperature changes, and the channel width's impact is considered infinitely large, ignoring curvature effects.

The highest volumetric flow rate achievable from the extruder, known as the drag volumetric flow rate shown in Figure 3, can be determined using the subsequent equation [28]:

$$Q_d = \frac{1}{2} \pi^2 D^2 N H \cos(\phi) \sin(\phi) \quad (1)$$

Where 'D' is the screw diameter, 'H' is the screw channel depth, 'N' is the screw rotational speed (RPM) and ' ϕ ' is the flight helix angle with vertical coordinate.

The maximum possible backward pressure is when the nozzle is closed. Accordingly, the maximum pressure can be calculated [28, 29]:

$$P_{max} = \frac{6\pi.L.D.N.\mu}{H^2.tan(\phi)} \quad (2)$$

Where 'D' is the screw diameter, 'H' is the screw channel depth, 'N' is the screw rotational speed (rpm), ' ϕ ' is the flight helix angle with vertical coordinate and ' μ ' is the viscosity of the paste.

Assuming Newtonian fluid behaviour, viscosity can be determined using the equation below, which factors in the shear rate imposed by the screw on the material:

$$\mu = m\gamma^{n-1} \rightarrow \mu = m\left(\frac{\pi.N.D}{60H}\right)^{n-1} \quad (3)$$

Where 'm' is the flow consistency index (or consistency coefficient), ' γ ' is the shear rate applied by the screw, 'n' is the flow behaviour index.

By employing the Hagen-Poiseuille equation [28-30], we can deduce the volumetric flow rate at the nozzle, expressed as follows:

$$Q(\Delta P) = \frac{\pi R^4 \Delta P}{8L\mu} \quad (4)$$

Where ' ΔP ' is the drop pressure at the nozzle, 'L' is the length of the nozzle, 'R' is the radius of the nozzle.

With the application of these equations, it's possible to construct a performance characteristic curve for the extruder. This graph, as shown in Figure 8, allows for the identification of operational points for various screw types. At such a point, the volumetric flow rate in the screw's metering section aligns with that at the nozzle. Thus, by adjusting the design parameters of the screw and nozzle, as well as the material properties – notably the viscosity – a stable output from the nozzle can be achieved [29].

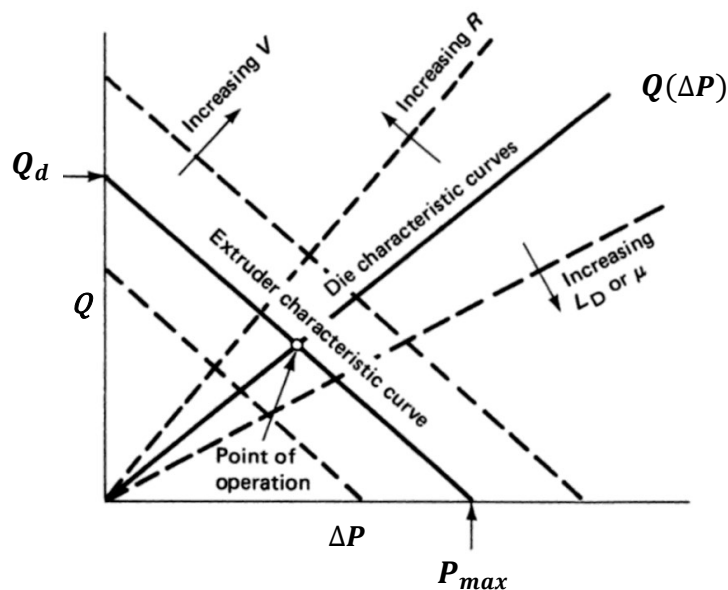


Figure 8. Performance characteristic curve for a simple viscous flow extruder: R = nozzle radius; L_D = nozzle-land length; μ = melt viscosity; L_c = length of screw channel; and $V = \pi DN \cos(\phi)$, the down-channel velocity at the barrel surface [29].

In integrating the extruder into a Hybrid Manufacturing (HM) system, a compact and lightweight design is essential for both research and industrial use. Consequently, screw and barrel diameters of 12, 16, and 20 mm, indicative of the smallest standard manufacturable sizes for three-zone screws, were assessed to determine the most suitable option for the system. Detail drawings of these screw geometries are available in the appendix.

Also, one key step involved identifying the smallest feasible nozzle diameter for each screw, crucial for the 3D prints' quality and nozzle size flexibility. The maximum extruder's viscosity capacity was set at 40,000 Pa.s at 0.3 s⁻¹ shear rate, surpassing conventional robocasting printers [1], thus enabling better processing of highly viscous materials.

A detailed spreadsheet, included in the appendix, was developed to assess three mentioned screw and barrel geometries, targeting a scenario where a screw design processes materials at 40,000 pa.s viscosity at 60 rpm through a minimum 0.3 mm nozzle while minimizing mechanical power.

Calculations of the drag flow output (Q_d) for each nominated screw geometry were performed using Equation 1. To establish the operational point for the extruder and nozzle, characteristic lines for both were plotted, as illustrated in Figure 9. The maximum pressure (P_{max}) within the screw, crucial for defining the operational range, was calculated using Equation 2. This process enabled plotting the extruder characteristic line, with key points at $(0, Q_d)$ for maximum flow rate with no nozzle and $(P_{max}, 0)$ when the nozzle is closed.

Additionally, by applying Equation 4 and varying pressure inputs (ΔP), a nozzle characteristic curve was developed, mirroring the approach shown in Figure 9. This curve, outlining the performance of the viscous flow extruder, aids in identifying the optimal operating points for various screw types. It specifically helps in aligning the volumetric flow rate in the screw's

metering section with that at the nozzle, ensuring efficient and consistent extrusion. An illustrative example is presented in Figure 9, which displays the performance characteristic curve for the 12mm screw diameter extruder, specifically emphasizing the scenario with a 0.3mm nozzle diameter.

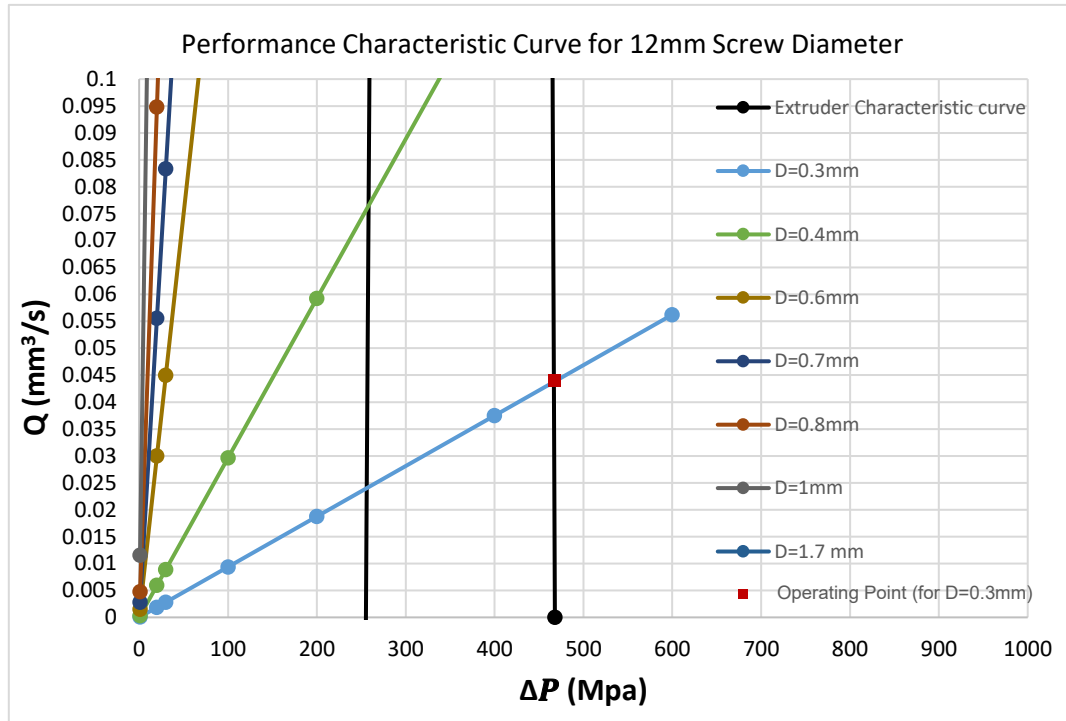


Figure 9. Performance characteristic curves for a 12mm screw diameter extruder, focusing on the worst-case scenario with a 0.3mm nozzle diameter. The graph illustrates the relationship between volumetric flow rate (Q , in mm^3/s) and pressure drop (ΔP , in MPa) for nozzle diameters from 0.3mm to 1.7mm. The highlighted curve for the 0.3mm nozzle and its operating point provides insights into the extruder's behaviour under the most demanding conditions, with comparative curves for larger diameters.

Table 1 presents a comparison of the operating point volumetric flow rates for three selected screw geometries, facilitating the selection of the screw that offers an acceptable flow rate and, consequently, the fastest extrusion capability. With the established operating point from Figure 9, the most stable flow rate for the selected screw can be predicted. Identifying this point allows for the subsequent calculation of the power and torque requirements necessary to maintain this operating condition.

Table 1 - provides a comparative overview of the operating flow rates for various screw diameters in a single screw extruder. It highlights that while the 16mm screw offers the highest flow rate, its viscosity range is notably more limited compared to the 12mm screw, emphasizing a trade-off between speed and material versatility.

Screw diameter (mm)	Total length (mm)	Min Viscosity (Pa.s)	Max Viscosity (Pa.s)	Volumetric Flow Rate of 0.3 mm nozzle at Max Viscosity (mm^3/s)	Compression ratio (ϵ)	L/D ratio
12	255	1000	40000	0.045	2.22	21.25
16	305	1000	20000	0.06	2.66	19
20	305	1000	70000	0.025	1.71	15.25

3.2.2. Required Power and Torque Calculation

The calculation of required torque and power for the designed screw is based on mechanical energy required method using shear stress concentration. This is demonstrated through the following equations [28-30]:

Power (P) is determined by:

$$P = F \cdot V \rightarrow P = \tau \cdot A \cdot V_{\text{Screw rotational speed}}$$

Leading to:

$$P = \frac{m\pi^{n+2}(D-2H)^{n+2}N^{n+1}L}{3600H^n \sin(\phi)} \quad (5)$$

Torque (T) is then calculated as:

$$T = \frac{P\pi}{60N} \quad (6)$$

Where 'm' is the flow consistency index (or consistency coefficient), 'n' the flow behaviour index, 'D' is the screw diameter, 'H' is the channel depth of the screw, 'N' is the screw's rotational speed, 'L' is the length of the screw, and ' ϕ ' is the helix angle.

These formulas establish the guidelines for choosing a suitable motor for the single screw extruder. For validation purposes, scaling methods are applied, adhering to standards like geometric uniformity in the design across extruders of different sizes, consistency in screw speed, and similar nozzle resistance. This strategy facilitates approximate scaling of power and torque needs, up or down, guaranteeing the extruder's design and functionality are both in an acceptable range [30].

$$P_2 = P_1 \left(\frac{D_2}{D_1}\right)^3 \quad (7)$$

$$T_2 = T_1 \left(\frac{D_2}{D_1}\right)^3 \quad (8)$$

Where the torques ' T_2 ' and ' T_1 ' and powers ' P_2 ' and ' P_1 ' of two extruders and their screw diameters ' D_2 ' and ' D_1 '.

To apply the Equations 5 and 6, a maximum rotational velocity for the extruder must be defined. Based on recommendations from screw and barrel manufacturers, a ceiling of 60 rpm, which applies 0.3 s^{-1} shear rate on material based on Equation 3, has been established as the upper limit for the extruder's speed. This rate was chosen to ensure the integrity and performance of the system. The remaining variables are determined by the specific geometry of the screw. Subsequent to the torque and power calculations for each screw geometry, a summary table was compiled, delineating the characteristics of each screw option—12, 16, and 20 mm in diameter—as presented in Table 2. This comparative table, derived from the comprehensive spreadsheet, facilitates an informed selection of the most efficient and appropriate screw for the intended application.

Table 2. Comparative Power and Torque Analysis of Screw Geometries for the Single Screw Extruder

Screw diameter (mm)	Total length (mm)	Min Viscosity (Pa.s)	Max Viscosity (Pa.s)	Volumetric Flow Rate of 0.3 mm nozzle at Max Viscosity (mm ³ /s)	Required Torque at Max Viscosity (N.m)	Required Power at Max Viscosity(w)	Compression ratio (ϵ)	L/D ratio
12	255	1000	40000	0.045	147	463	2.22	21.25
16	305	1000	20000	0.06	215	677	2.66	19
20	305	1000	70000	0.025	323	1017	1.71	15.25

Table 2 reveals that the 12 mm screw diameter can extrude materials with the highest viscosity while maintaining a satisfactory volumetric flow rate at the determined operating point, requiring the least amount of energy and torque. Furthermore, its length-to-diameter (L/D) and compression ratios align with recommendations from the literature review. Consequently, the 12 mm diameter screw has been selected for the extrusion component of the hybrid manufacturing machine due to its efficiency and compatibility with established guidelines.

Table 2 indicates that the 12 mm screw necessitates a power (P) of 463 watts to apply a torque (T) of 147 N.m, enabling the extrusion of a material with 40,000 Pa.s viscosity through a 0.3 mm nozzle at a rotational speed (N) of 60 RPM, identified as the extruder's most challenging scenario. Factoring in an efficiency rate of 40% for the power system, the total motor power required for extrusion is calculated to be 1.2 kW. To validate these calculations and the power requirement, Equations 8 and 9 were utilized. Given the lack of publicly available specifications for many industrial extruders, a scaling down of comparable industrial injection molding equipment with analogous geometry was employed to validate the computed power and torque figures. Utilizing data from Shibaura Machine Company's single screw injection extruder [36] equipped with a 37kW motor and a 50mm screw diameter, and applying Equation 7, the required power for an extruder with a 12mm screw diameter is calculated to be approximately 600W. This data confirms that a 1.2kW motor will suffice for the extruder.

In response to the project funder AM-COE's requirement for a machine capable of adapting to larger screw geometries, a 1.2kW motor was selected. Following a similar analytical approach, a reduction gearbox was also chosen to enhance the motor's torque capacity to 160 N.m, ensuring the system's suitability for a wider range of applications.

3.2.3. Extruder Motor Type and Gearbox Selection

In the design of the Hybrid Manufacturing machine, the selection of an AC servo motor with an integrated electromagnetic brake is essential due to its high torque and precise position control. This motor is chosen for its closed-loop control system, which ensures accurate torque regulation and positioning—critical for consistent extrusion of varying-viscosity materials. The motor's rapid response and efficiency are crucial for maintaining high precision in manufacturing.

The addition of an integrated brake enhances system safety and control, enabling immediate halts in operation to quickly address any process discrepancies. This feature minimizes downtime and facilitates rapid adjustments, ensuring continuous and reliable production.

The chosen motor provides a maximum torque of 12 N.m and a rated torque of 4 N.m. It is paired with a planetary gearbox with a 40:1 reduction ratio, boosting the torque to 160 N.m, necessary for processing high-viscosity materials. This combination ensures consistent, powerful extrusion performance, underpinning the machine's capability to produce high-quality ceramic components efficiently.

3.2.4. Screw and Barrel Material Selection

The selected material for both the screw and barrel is 38CrMoAl, a high-grade alloy structural steel with a yield strength of 850 MPa. This material is commonly employed in the industry for fabricating screws and barrels of extruders due to its robustness and durability.

Figure 8 illustrates that in the worst-case scenario, the extruder is subjected to a pressure of approximately 500 MPa. Applying a safety factor of 1.5, the resulting stress is about 750 MPa, which is comfortably below the 850 MPa yield strength of 38CrMoAl. This margin ensures the reliability and structural integrity of the screw and barrel under operational conditions.

3.2.5. Final Technical Extruder Specification

Table 3. Extruder System Specifications for Hybrid Manufacturing Machine

Parameter	Value
Weight	25 kg
Material Viscosity Range at 0.3 s^{-1}	1000 – 40000 pa. s
Nozzle Diameter Size Range	0.3 – 1.7 mm
Min Filament Extrusion Speed	5.6 – 200 mm/s
Volumetric Flowrate (Varying according to material and nozzle size)	$0.06 - 215 \text{ mm}^3 \text{ s}^{-1}$
Operational Rotational Screw Speed	Up to 60 RPM
Motor power	1.2kW
Planetary gearbox ratio	40
Max temperature	400 °C
Max environment temperature	60 °C
Compression ratio	2.2
L/D ratio	21.25
Screw Diameter	12 mm

3.3. X-Y-Z Axes Design and Implementation

The design of the X-Y-Z axes is critical for the precision and efficiency of the Hybrid Manufacturing (HM) machine. To this end, linear stages equipped with ball screws have been utilized across all three axes due to their exceptional positional accuracy and reliability. The

configuration includes two parallel X-axes with a travel range of 900 mm, dual back-to-back Y-axes with a 600 mm stroke, and two Z-axes each providing a 300 mm stroke, ensuring expansive operational flexibility.

Each axis of the HM machine is powered by an AC servo motor, chosen for their smooth, precise control and silent operation. These motors enable a closed-loop system essential for the detailed work of additive and subtractive manufacturing. Their programmable nature provides flexibility, adapting easily to diverse production requirements. This precision positions the HM machine as a leader in the field, ensuring the highest accuracy in crafting complex ceramic components and elevating the quality and consistency of the outputs.

3.4. X-Axis Printing Bed Assembly

The design of the X-axis printing bed assembly, refined through "Initial Tests and Proof of Concept," incorporates a crucial heated platform to enhance first-layer adhesion, essential for part stability throughout both printing and CNC machining. A 24V Aluminium heating element, chosen for its efficiency in rapid and uniform heating, accommodates various bed coverings to suit different materials.

Spring-mounted with manual levelling knobs, the bed's construction prevents undesired heat transfer to sensitive components and guarantees level printing aligned with the extruder's X-Y movements. This design is critical in averting material warping, thus preserving the geometrical integrity of the parts. With dimensions of 25 x 25 mm, the bed offers ample space for a wide range of part sizes, underpinning the assembly's versatility in the HM machine's robust manufacturing capabilities.

3.5. Structural Body Design

As delineated in Figure 4 (a), the structural framework of the HM machine utilizes four 160x80 aluminium profiles to support the Y-axis, upon which the extrusion assembly are mounted. Aluminium profiles were selected for their versatility in accommodating fasteners, facilitating ease of assembly—crucial for the initial prototype of the HM machine. Additionally, these profiles provide the requisite strength for the machine's overall structure, balancing durability with flexibility.

Atop these columns, two 40x40 aluminium profiles are conjoined to act as a bridge, reinforcing the structural integrity between the columns while also serving as a support for the Y-axis linear guides. This arrangement enhances the stability and precision of the movement along the Y-axis.

The entirety of the machine rests on a 1 m by 1.7 m granite base. This foundation not only isolates the machine from environmental vibrations but also contributes significantly to its damping capabilities, courtesy of the granite's substantial weight of 800 Kg. Moreover, granite offers exceptional flatness, a quality indispensable for achieving the high precision requisite in the HM machine's operation. This thoughtful integration of materials and structural design

ensures that the HM machine stands robust against operational stresses while maintaining the precision and stability needed for advanced manufacturing processes.

3.6. HM Machine Control Systems

This section will articulate the operational logic and architecture of the control systems integral to the HM machine, employing a hierarchical approach from high-level concepts to detailed component interactions.

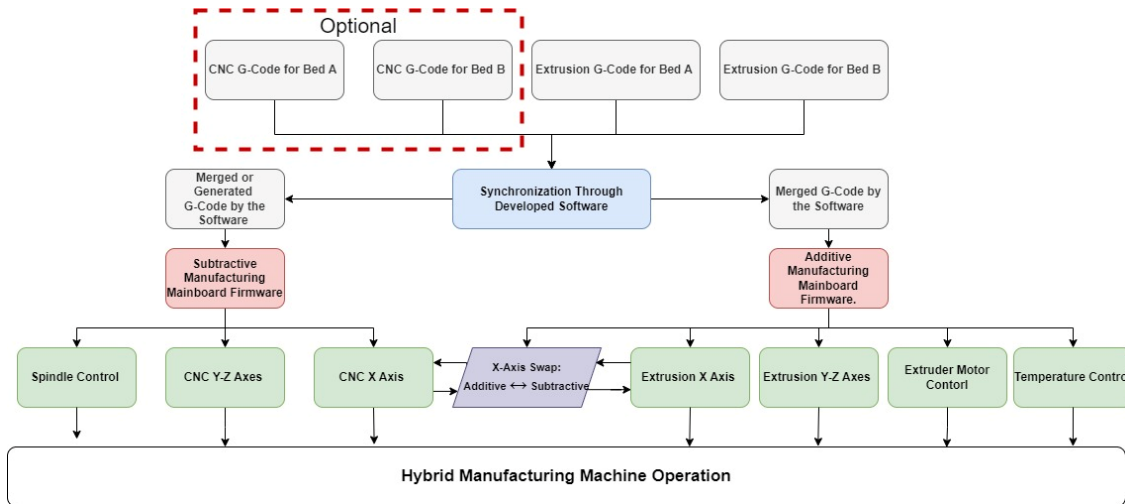


Figure 10 - High-Level Control Flow Diagram of the Hybrid Manufacturing Machine, detailing the harmonized interaction between slicing software, mainboard firmware, and the synchronization software. Initially, the slicing software generates four G-code files—two for CNC machining and two for robocasting 3D printing, each corresponding to separate beds (A and B) within the machine. These G-codes are crucial as they contain the necessary instructions for manufacturing the components on each bed.

The machine utilizes two distinct mainboard firmware systems to independently manage the subtractive and additive manufacturing processes. The Additive Manufacturing Mainboard Firmware regulates the 3D printing operation, controlling the temperature, the extruder motor, and the XYZ axes. On the other side, the Subtractive Manufacturing Mainboard Firmware oversees the spindle motor and the XYZ axes for CNC machining. Each mainboard operates based on the G-code provided for its respective bed, ensuring precise control over each manufacturing section.

A specialized synchronization software acts as a central command hub, receiving all four initial G-codes. It meticulously merges and orchestrates these into two comprehensive G-codes—one for the subtractive and one for the additive mainboard. This innovative software not only facilitates synchronization but can also generate the CNC G-code tool paths from the 3D printing G-code for the subtractive process, an optional feature that simplifies the transition from additive to subtractive operations. This optional capability, while not extensively explored in this report, represents a significant step forward in operational efficiency.

The synchronization software ensures the additive and subtractive mainboards are precisely aligned, enabling the machine to execute layer-by-layer CNC machining of 3D printed structures. As a result, this synchronized operation gives rise to the Hybrid Manufacturing

Machine's operational flow, where additive and subtractive manufacturing are seamlessly integrated to produce high-precision components with improved surface finishes.

Simultaneously, the machine maintains rigorous control over other essential parameters. Temperature control and motor management across all axes are tightly regulated, allowing for a synchronized and stable manufacturing environment. This level of control is vital in achieving the precision required in advanced manufacturing, and the HM machine's design ensures that these parameters are meticulously observed throughout the manufacturing process.

3.6.1. Mainboard Controller and Firmware

In the realm of hybrid manufacturing, the choice of a mainboard controller is crucial for precision, stability, and flexibility. The Duet 3 Mainboard 6XD stands out for its robust power delivery tailored for 3D printing, its proven reliability, and system stability, making it the mainboard of choice for the extruder section of this machine. It hosts the RepRap firmware, an open-source, highly customizable firmware, which has been instrumental in this project. Its customizability, facilitated by access to source code, has allowed for significant adaptations, including the integration of CNC machining capabilities—a feature indispensable for a hybrid manufacturing setup.

The board's web interface is another boon, offering user-friendly access to machine controls via a wireless connection. It extends the capabilities of the machine, promoting user interaction and facilitating remote management. Figures 11 and 12 display the two control options for the additive manufacturing part of the Hybrid Manufacturing machine: a web-based interface for wireless access and a touch screen interface, allowing users to select the most suitable method for their needs.

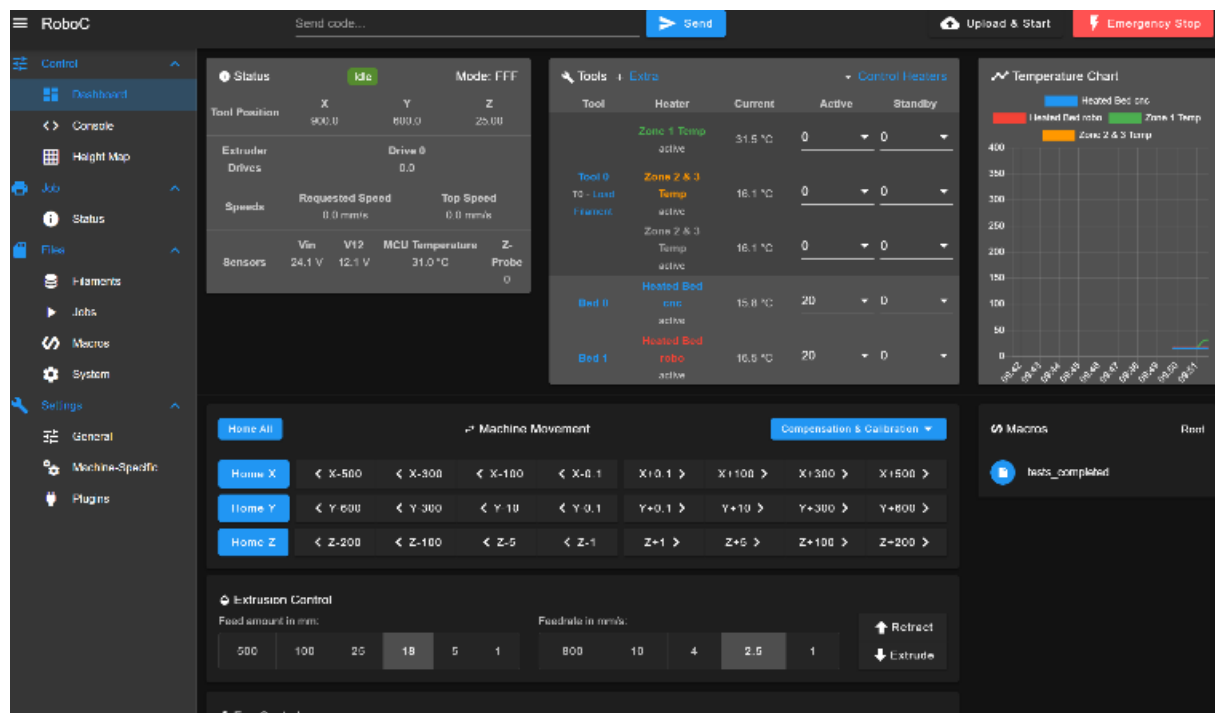


Figure 11 - User Interfaces of the Hybrid Manufacturing Machine, Web user interface accessible via wireless connection across the LAN network.

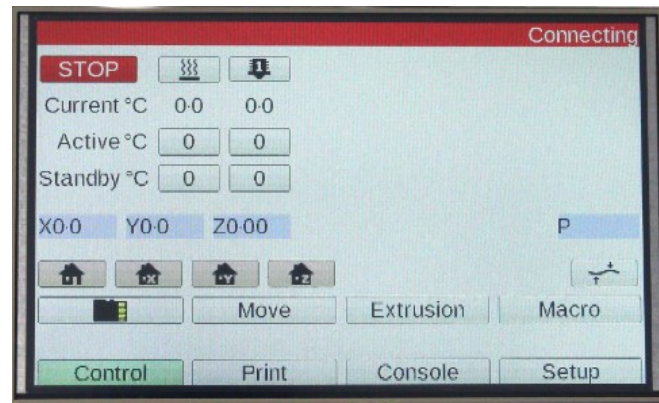


Figure 12 - LCD user interface, offering users a choice of operational control methods.

Furthermore, the board's versatility in generating pulse and sign signals for AC servo motor control, along with its assortment of PWM, IO, 24V, and 12V pins, confirms its suitability for the complex demands of the HM machine. The wiring schematic and detailed connections highlighting these features are illustrated in Appendix, providing a clear visual guide to the intricate electronic orchestration that powers the machine.

3.6.2. Temperature Control

The Temperature Control section is pivotal, especially given the temperature sensitivity of the extrusion materials and their properties, such as viscosity and flowability. To ensure comprehensive temperature management, various sensors are strategically deployed throughout the extruder.

Extruder Temperature Management:

In the extruder assembly, two K-type thermocouples are employed to monitor temperatures across compression and metering zones. One thermocouple monitors the feeding zone temperature, guiding the heater in Zone 1, while another regulates the heaters in the compression and metering zones, Zones 2 and 3, for accurate temperature control. The heaters—220V brass heat bands with a 200W capacity—provide ample and stable heat over time. A PID controller regulates these temperatures, ensuring stability and minimizing system delays. Individual AC-DC solid-state relays (SSR) control these heaters, allowing for precise activation and deactivation orchestrated by the main control system.

Feeding Section Cooling:

For cooling, a 12V GDSTIME fan with a capacity of up to 5000 RPM and 103.4 CFM airflow is integrated into the feeding section. This fan, managed by PWM signals from the mainboard, responds to the thermal readings from the first zone's K-type thermocouple sensor. The positioning of the fan ensures effective cooling of both the feeding section and the adjacent heatsinks, maintaining the desired temperature gradient within the extruder.

Heated Bed Temperature Control:

The heated beds of the HM machine utilize 24V Aluminium heating elements for rapid and uniform heat distribution, adaptable to various bed coverings. Temperature monitoring is achieved through thermistors, providing precise surface readings. Each bed is managed by a PID controller linked to a solid-state relay, ensuring exact temperature regulation. This effective temperature control system is essential for maintaining steady bed temperatures, crucial for consistent print quality and ensuring printed layers remain securely attached to the heated bed during CNC machining.

3.6.3. X-Y-Z Motor Controls for Precision Manufacturing

For the tri-axial (XYZ) movement within the HM machine, AC servo motors were selected for their superior position accuracy, robust torque, and the flexibility provided by programmable drivers, thereby establishing an efficient closed-loop position control system. Each axis is precisely controlled via a PID positional control, finely tuned within the motor drivers. To ensure seamless integration with the Duet 3 MB 6XD mainboards, the motor drivers are set to operate in 5V pulse and sign signal position control mode, allowing for exact manipulation based on the board-generated signals. Additionally, a brake is integrated into the Z-axis motor, which activates when the motor is disabled. This feature is crucial for maintaining the position of the heavy extruder assembly attached to the Z-axis, ensuring it remains securely in place without falling. The specifications and detailed drawings of these selected AC servo motors are provided in the Appendix.

Optical limit switches are utilized for the X-Y axes homing, chosen for their reliability, while the Z-axis features a BLTouch probe that handles both homing and automatic levelling. This integration allows the machine to dynamically map the print bed's surface, adjusting the G-code to accommodate variations in bed height. These adjustments enhance the print quality by ensuring proper alignment and consistent layer deposition across the print bed [37].

3.6.4. Extruder Motor Control

The extruder's AC servo motor features an integrated electromagnetic brake, crucial for controlling operational precision. Activated and controlled by G-code commands whenever the motor stops, the brake swiftly halts movement, enhancing print quality and reducing system delays. It effectively handles the high torque of the extruder material, ensuring stability and preventing unwanted movement during operational pauses.

Following the mechanical safety provided by the brake, the motor's driver can be programmed for torque or position control. It operates within an analogue command voltage range of -10V to 10V. To implement the closed loop torque control, a Digital-to-Analog Converter (DAC) is employed to convert pulse and sign signals from the Duet 3 MB 6XD board into the necessary analogue commands for the motor driver. Additionally, the motor can be operated using position control, managed directly by pulse and sign signals, providing an alternative control mode that enhances precision. This dual capability ensures that the motor's response is

meticulously aligned with the specific requirements of the extrusion process, facilitating smooth and consistent material flow.

3.6.5. Synchronization Hardware

To ensure seamless synchronization between the extrusion X-axis and the CNC X-axis, a dedicated synchronization method has been developed. This method involves using a DPDT (Double Pole Double Throw) relay that switches the Pulse and Sign signals controlling the X-axis motors of both the subtractive and additive systems. By swapping these signals, control is seamlessly transferred between the subtractive and additive controllers, facilitating the alternating use of the two beds.

However, synchronization is not just about switching control; it also involves ensuring that each controller is aware of the exact position of the beds at the time of swapping. To expedite this process, instead of re-homing or recalibrating the beds, the system adopts a more efficient approach. For each swapping sequence, a developed software takes the last known X-axis position from the additive mainboard's G-code and directly sets this as the starting position in the subtractive mainboard's sliced G-code and vice versa. This strategy significantly speeds up the synchronization process, reducing downtime and enhancing the overall efficiency of the machine's operation. This calibration method is managed within the developed synchronization software, which is detailed in the subsequent section 3.6.6.

3.6.6. G-Code Generation & Synchronization Software

The synchronization process within the Hybrid Manufacturing Machine commences when the user utilizes Cura, or other open-source slicing software, to generate two sets of G-code files for additive manufacturing, one for each printing bed, Bed A and Bed B. In parallel, two additional G-codes for the subtractive CNC machining processes are created using open-source CAM software specific to the parts designated for each bed. These four G-codes—two for additive manufacturing and two for CNC machining—are then processed by a synchronization software developed as part of the machine's integrated system.

This advanced software, diagrammed in Figure 10, intelligently combines the separate G-codes into two harmonized instruction sets. One set commands the Additive Manufacturing Mainboard Firmware, while the other directs the Subtractive Manufacturing Mainboard Firmware. The innovative software design incorporates a user-defined Machining Per Layer (MPL) feature, which specifies the frequency of CNC machining relative to the number of layers printed. For instance, setting the MPL to 1 would result in layer-by-layer machining, enabling the user to decide after how many layers of additive manufacturing the CNC machining should intervene. This flexibility is crucial for tailoring the manufacturing process to the specific needs of the component being produced.

The synchronization software also features an optional tool path generation for the CNC machining based on the 3D printing G-code, mitigating the need for separate CAM software if so desired. Tool and nozzle specifications are input through the user interface, shown in Figure 13, further customizing the generated G-code for CNC machining. This function represents an

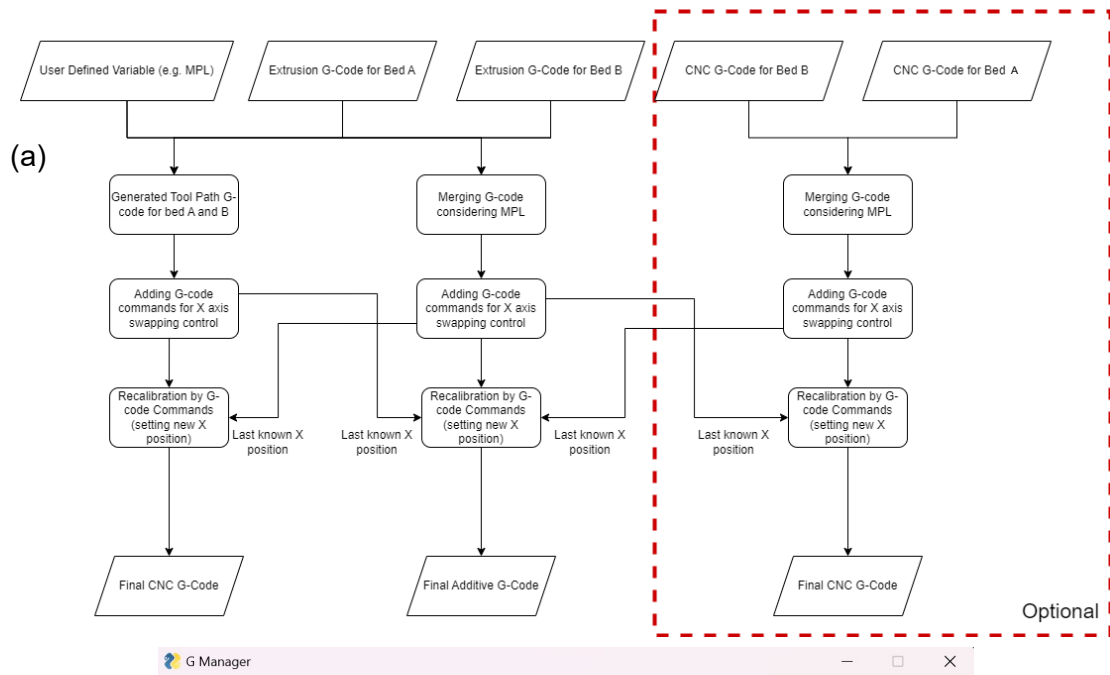
optional enhancement that can streamline the transition from printing to machining, though it has not been the primary focus of this report.

Additionally, the software plays a critical role in coordinating the control systems. It generates commands to activate the DPDT relay explained in section 3.6.5 to facilitate the swapping of the X-axis control between additive and subtractive mainboards.

As the beds swap, the software accurately recalibrates the position of the X-axis by implementing G-code commands generated for both the additive and subtractive mainboards. This process ensures that each mainboard is precisely informed of the X-axis's position at the moment of the swap. Due to the common reference point for both parallel X-axes, the position remains calibrated and consistent across operations. By maintaining this level of synchronization, the software enables a smooth transition between stages, ensuring that the manufacturing process continues with precision.

The ingenuity of the synchronization software lies in its ability to ensure that the additive and subtractive mainboards effectively communicate and wait for each other's completion of tasks before proceeding with the bed swapping. This coordination is critical for maintaining the integrity and efficiency of the machine's operations, positioning the Hybrid Manufacturing Machine as a reliable platform for high-precision manufacturing.

In summary, this synchronization software is integral to the operation of the Hybrid Manufacturing Machine, adeptly merging G-codes and managing synchronization to facilitate smooth and efficient transitions between additive manufacturing and CNC machining.



(b)



Additive G-code File 1:		Browse
Additive G-code File 2:		Browse
Subtractive G-code File 1:		Browse
Subtractive G-code File 2:		Browse
Precise Additive G-code:		Browse
Select Output Folder:		Browse
MPL		
Spindle Speed (RPM):		
Feed Rate (mm/min):		
Tool Length (mm):		
Tool Diameter (mm):		
Nozzle Diameter (mm):		
Safety Factor:		
Ex Layer Thickness (mm):		
Submit	Cancel	

Figure 13 - (a) Control diagram illustrating the operational logic of the custom-developed software. (b) User interface of the software, capable of generating G-code commands tailored to user specifications for simultaneous additive and subtractive manufacturing operations.

3.7. Design of the Experiments (DoE)

To response to aims and objectives of this project, two experiments are designed to validate the outcome of the project.

3.7.1. Position Accuracy XY Experiment

To validate the initial aim of demonstrating the positional accuracy of the X-Y axes, a targeted experiment has been conducted. For this test, a pen was attached to the extruder in place of the usual printing head. The controller was then programmed with the G-code for drawing a circle. The STL file used specified a circle with a radius of 12.5 mm, ensuring the experiment's parameters were precisely defined to assess accuracy.

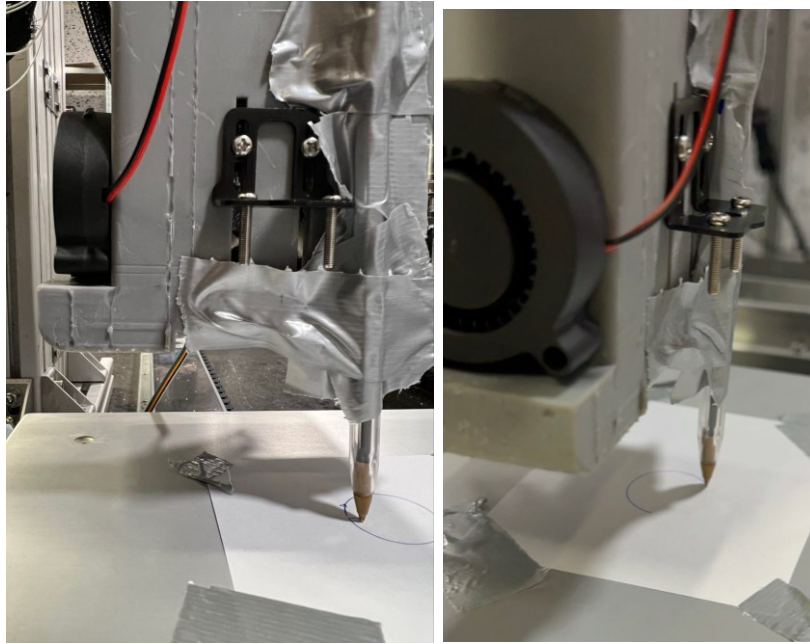


Figure 14 - Positional Accuracy Test of the Hybrid Manufacturing Machine demonstrating a pen attached to the extruder assembly drawing a circle with a predefined radius on paper. This experiment is designed to validate the precision of the X-Y axes movement.

3.7.2. Temperature Stability Experiment

The Temperature Stability Experiment is designed to validate the functionality of the temperature control system within the extruder and heated beds of the Hybrid Manufacturing machine, ensuring they operate within the specified design parameters. The experiment assigns specific temperatures to each component involved in the extrusion process, as detailed in Table 4.

The procedure involves setting these temperatures and continuously monitoring them with integrated sensors. The data collected from these sensors are then used to assess the effectiveness of the PID controllers in maintaining the designated temperatures throughout the operation.

This data will be analysed by plotting it on a PID response curve, allowing for an evaluation of the temperature control's precision across the machine's various sections. This experiment is crucial for confirming that the machine's material processability and overall reliability meet the project's design requirements.

Table 4 - Set temperatures for the Hybrid Manufacturing machine's components, including Heater Zones 1, 2, and 3, and Heated Beds A and B.

Heater Name	Set Temperature (°C)
Heater Zone 1	150
Heater Zone 2 and 3	150
Heated Bed A	50
Heated Bed B	50

3.7.3. Extrusion Capability Experiment

This experiment comprises two distinct phases, each aimed at assessing the Hybrid Manufacturing machine's capacity to extrude high-viscosity thermoplastic pellets provided by AM-COE. The material's viscosity at 180 °C falls well within the operational capabilities of the extruder, as delineated in the "3.2.5. Final Extruder Specification" section. This alignment ensures that the material characteristics are compatible with the extruder's specifications, setting the stage for a thorough evaluation of the machine's extrusion performance. The first part focuses on assessing the repeatability and reliability of the extruded filament, while the second part examines the machine's proficiency in forming a precise three-dimensional object.

Experiment 1: Filament Extrusion Reliability and Repeatability

In the initial phase of the experiment, thermoplastic pellets are manually loaded into the hopper and introduced into the extruder. The machine, equipped with a 0.8 mm nozzle, processes the pellets into a continuous filament. To assess the effectiveness of the die swelling fans, the experiment is conducted twice—once with the die fans turned on and once with them off, as illustrated in Figure 7(a). The uniformity and consistency of the extruded filament are meticulously evaluated by measuring its thickness at various intervals using a calliper. This setup tests the extruder's ability to perform consistently, and the reliability of the filament produced, ensuring that the machine can sustain stable extrusion conditions throughout extended operations.

Experiment 2: Three-Dimensional Precision Extrusion

The second phase involves extruding the same thermoplastic pellets into a predefined three-dimensional shape—a 10x10x10 mm cube. An STL file for the cube is prepared and sliced using Cura, set to a normal profile with a 0.2 mm layer thickness. Post-extrusion, the cube is examined under an electron microscope to measure the layer thickness and the dimensions of each side (X, Y, and Z) to assess the accuracy against the intended dimensions.

The chosen geometric forms, particularly the simple square, are instrumental in demonstrating the extruder's capability to handle and precisely deposit AM-COE material. The primary objective is to validate the extruder's fundamental functionality against the project's design specifications without adjusting the printing parameters for optimization. Successfully producing uniform filament and precisely shaped cubes will substantiate the extruder section's design efficacy, confirming its capability to meet the specific demands of high-viscosity

ceramic-thermoplastic material extrusion. This experiment not only provides a proof of concept for the extrusion system but also reinforces the reliability and precision of the Hybrid Manufacturing machine within a controlled experimental setup.

3.7.4. Hybrid Manufacturing Synchronization Experiment

In the synchronization capability assessment of the Hybrid Manufacturing machine, the methodology mirrors the settings and parameters of the 3.7.3 Extrusion Capability Experiment 2. Under the same conditions—temperature settings, nozzle diameter of 0.8 mm, and material specifications—a 10x10x10 mm cube is first printed on Bed A, drawing on the machine's additive manufacturing capabilities.

Upon the successful deposition of the cube, the experiment transitions to the synchronization test, a critical evaluation of the machine's integrated functionality. The synchronization process, including bed swapping and CNC machining initiation, is orchestrated by the developed software, as detailed in section 3.6.6. The software ensures the precise execution of the necessary G-code commands, derived from the user-defined variables, to commence the automated transition between the additive and subtractive stages.

As the first cube undergoes CNC machining after the swap, a second cube's printing begins on Bed B. This simultaneous operation is not just a display of the machine's multitasking ability but is a fundamental test of the synchronization between the two distinct manufacturing processes. The synchronization's success would illustrate the machine's capability to smoothly transition between these processes without manual intervention, signifying a robust proof of concept.

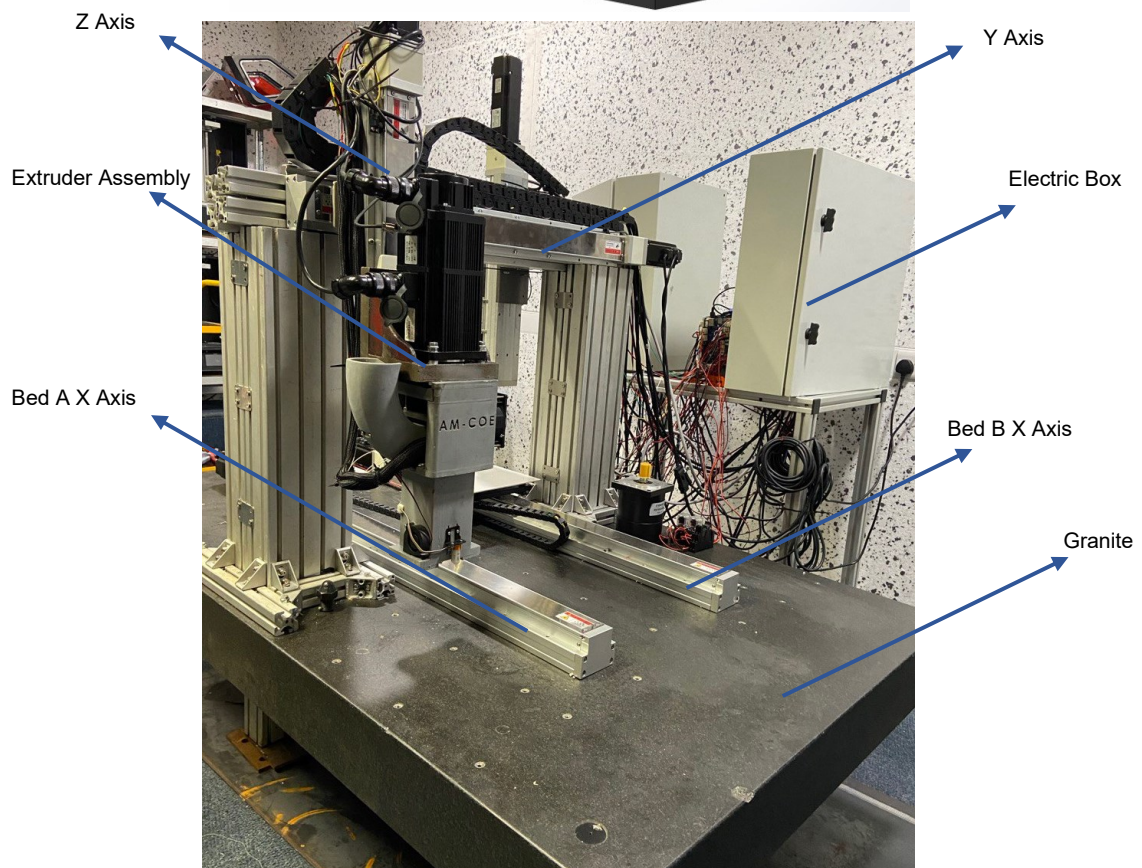
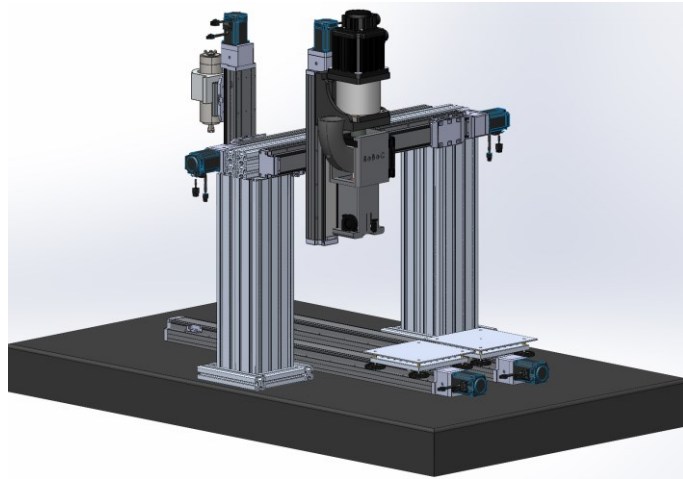
This experiment is paramount in demonstrating the practical application of the machine's designed synchronization system, highlighting the seamless integration of additive and subtractive manufacturing. The objective extends beyond mere functionality; it aims to respond to defined aim of this project and establish the Hybrid Manufacturing machine as a viable solution for advanced manufacturing needs where the combination of additive layer creation and precise CNC finishing is crucial.

4. RESULT AND DISCUSSION

The culmination of the Hybrid Manufacturing (HM) machine project presents tangible results that reinforce the research objectives and validate the machine's capabilities. The physical manifestation of the HM machine stands as a testament to the feasibility and practicality of integrating additive and subtractive manufacturing processes. The following segments illustrate the outcomes of the developed machine and the experiments conducted to test its functionality.

4.1. Fabrication and Assembly of the HM System

The HM machine, as constructed, embodies the envisioned design, marrying robustness with precision. Figure 15 showcases the actual built machine in its operational environment. The image captures the juxtaposition of the additive and subtractive sections and highlights the machine's comprehensive structure and the integration of its key components. This real-world build confirms the design's viability and the successful translation from concept to reality.



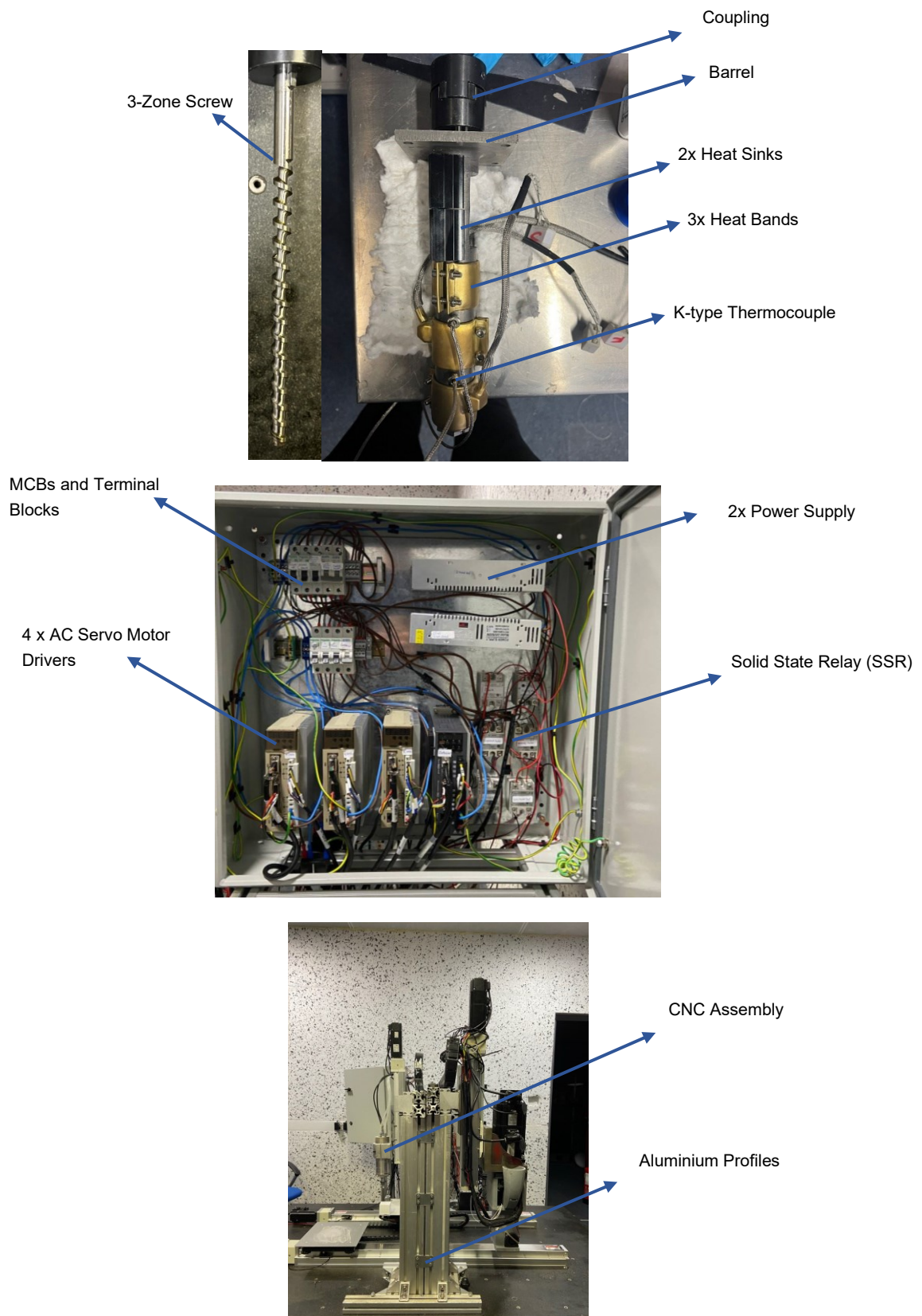


Figure 15 - Actual build of the Hybrid Manufacturing Machine, highlighting the integration of additive and subtractive manufacturing components, and electrical box including all wiring for this project.

4.2. Results of Conducted Experiments

4.2.1. Position Accuracy XY Experiment

Following the experiment designed in section 3.7.1, the Position Accuracy XY Experiment assessed the X-Y axes control of the extruder section. The test involved measuring the internal and external radii of a circle drawn by the machine, with results showing an internal radius of 12.34 mm and an external radius of 12.67 mm. Calculating the average of these measurements yielded a central radius of 12.505 mm, exhibiting a nominal deviation of only 5 microns from the set target of 12.5 mm. These findings are documented as the first trial in Table 5 and visually represented in Figure 16.

Table 5 - Positional Accuracy Measurement Results: Detailed results from five trials measuring the internal and external radii of a circle drawn by the extruder section, including calculated central radii, deviations in micron, and percentage errors. This table demonstrates the consistent precision of the X-Y axes control system.

No. Trial	Internal radius (mm)	External radius (mm)	Central radius (mm)	Central radius error (micron)	Central radius error (%)
1	12.34	12.67	12.505	5	0.04
2	12.337	12.665	12.501	1	0.008
3	12.338	12.67	12.504	4	0.032
4	12.337	12.67	12.5035	3.5	0.028
5	12.34	12.65	12.495	-4.9	-0.04

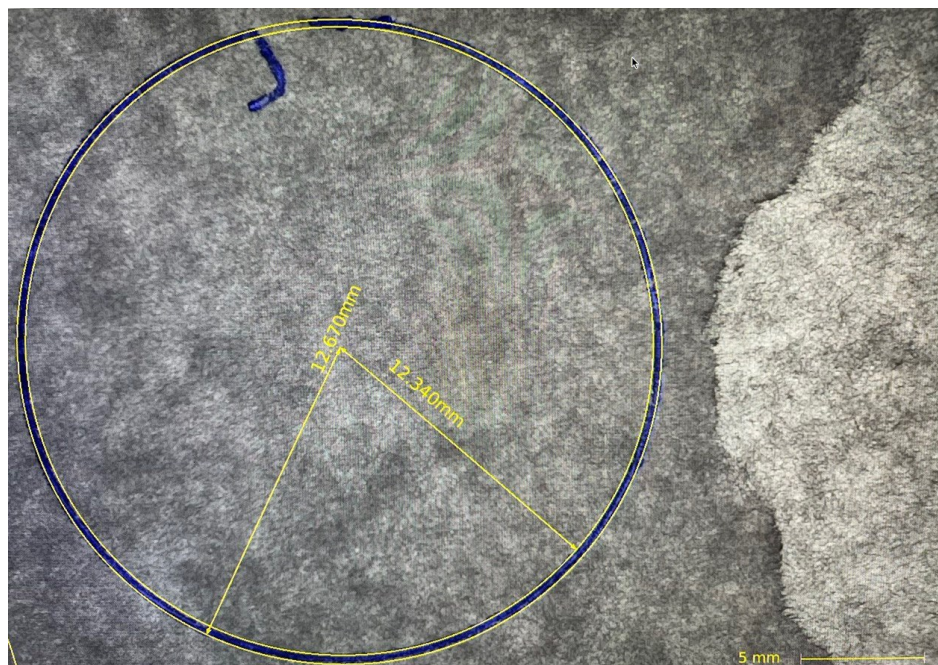


Figure 16 - Electron microscope image of a precision test circle drawn by the HM machine, showing internal and external radii measurements of the first trial and a central deviation of ± 5 micron.

Table 5 illustrates the HM machine's X-Y positional accuracy, showcasing deviations from 0.008% to 0.04%. This consistent precision, held within ± 5 micron, underscores the effectiveness of the extrusion section's design and the integrated closed-loop PID control system in the AC servo motor drivers. Such accuracy adheres to the project's rigorous design criteria, affirming the HM machine's reliability and operational performance.

To further validate these results and address potential variability introduced by the experimental setup—such as the pen attachment which might cause slippage on the paper—a more precise measurement method like a laser tracker could be used in future tests. This would refine the accuracy assessments by minimizing external mechanical influences.

This integration of data and visual evidence effectively communicates the success of the project's objectives, showcasing the HM machine's potential in precise X-Y position accuracy within a controlled experimental framework.

4.2.2. Temperature Stability Experiment

The Temperature Stability Experiment, delineated in section 3.7.2, rigorously tested the accuracy and stability of the temperature control system within the Hybrid Manufacturing machine. The assessment focused on the extruder's heaters and heated beds, each assigned specific target temperatures as documented in Table 4.

Heater Zone 1, responsible for the feeding section and set at 150°C, reached this temperature within 90 seconds and stabilized after 12 minutes. Heater Zones 2 and 3, catering to the compression and metering zones with the same temperature setting, attained their operational temperatures within 90 seconds and required 300 seconds to stabilize. The control system for these heaters is intentionally designed to be underdamped, enabling rapid heating to reach target temperatures swiftly, thereby accelerating the warming of the extruder barrel. It's important to note that the entire machine requires at least 20 minutes to stabilize and be ready for operation. This preparation time falls within the expected range for machine readiness in control engineering terms. Heated Beds A and B, both set to 50°C, were designed to be critically damped and maintained their designated temperatures after only 60 seconds. These components demonstrated exceptional thermal stability over a continuous nine-hour period, with temperature fluctuations confined within a precise ± 2 degrees Celsius.

The data collected was illustrated on a PID response curve (see Figure 17), which captures the dynamic stability and response rates of the temperature control system. This graphical representation highlights the system's effectiveness in rapidly achieving and consistently maintaining the set temperatures across all zones achieving the project's aim.

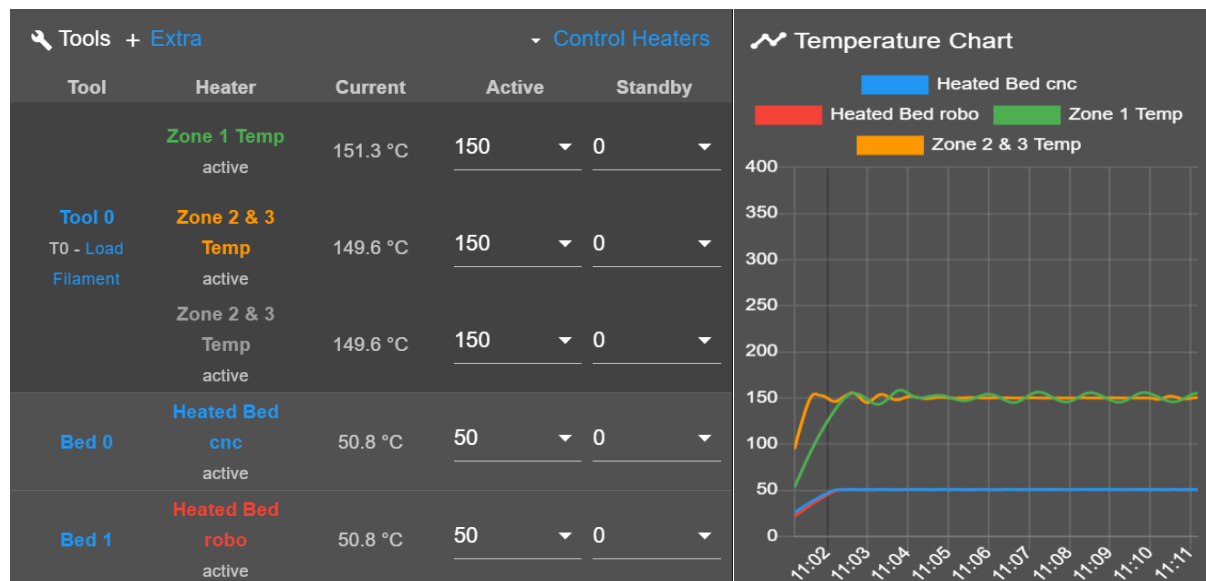


Figure 17 - PID response curve illustrating the temperature stability over time for Heater Zones 1, 2, and 3, and Heated Beds A and B.

Despite the successful outcomes, the ± 2 -degree Celsius fluctuation observed suggests a potential area for further refinement, particularly in the tuning of the PID controllers for Heater Zones 1, 2, and 3. Optimizing these settings could enhance the system's overall thermal stability. Nonetheless, the performance metrics achieved have been ratified by AM-COE and align with the fundamental operational requirements of the extrusion process, validating the temperature control system's capability to support advanced manufacturing applications effectively.

4.2.3. Extrusion Capability Experiment

The results of the Extrusion Capability Experiment underscore the Hybrid Manufacturing machine's adeptness in managing and extruding high-viscosity thermoplastic materials supplied by AM-COE. Both experimental phases aimed to rigorously assess the extruder's performance in terms of repeatability, reliability, and dimensional accuracy.

Experiment 1: Filament Extrusion Reliability and Repeatability

The primary stage of the experiment was designed to ascertain the Hybrid Manufacturing machine's proficiency in producing a uniform thermoplastic filament from AM-COE supplied pellets. Throughout this phase, the filament's thickness was meticulously measured at intervals along its length, which is depicted in Figure 18. In the ideal state, the filament's diameter should consistently match the nozzle size of 0.8 mm.

Subsequent analysis indicated an average deviation of 73 microns and maximum error of 130 microns from the ideal diameter across ten trials without the die cooling fans engaged. This variance is notably depicted in Figure 19, where the resulting data points underscore the filament's inconsistent diameter. However, upon activation of the die cooling fans, a significant enhancement in uniformity was observed. The average error contracted sharply to around 10 microns and maximum error of 30 microns, a testament to the cooling fans' efficacy in

managing the extrudate's temperature and mitigating die swell phenomena. Figure 19 illustrates the reduction in filament thickness variation, suggesting a correlation between the cooling duration and dimensional stability.

This result not only confirms the reliability of the die cooling system but also underscores the extruder's capability to ensure consistent output, a critical factor for high-volume production scenarios. This evidence of improved precision with the integration of cooling measures affirms the extruder's ability to adhere to strict manufacturing tolerances.

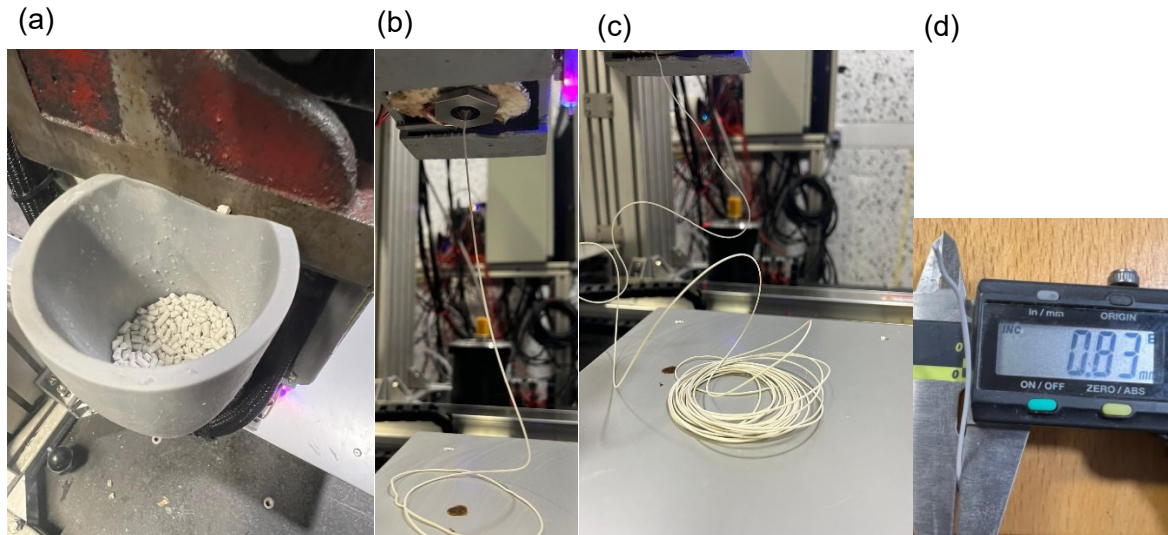


Figure 18 – Key stages in the extrusion process for filament consistency testing. (a) displays AM-COE's ceramic-thermoplastic pellets loaded in the hopper; (b) shows the material being extruded through a 0.8 mm nozzle; (c) captures the extruded filament coiling for measurement; and (d) illustrates the use of a calliper for precision thickness measurement along the filament's length.

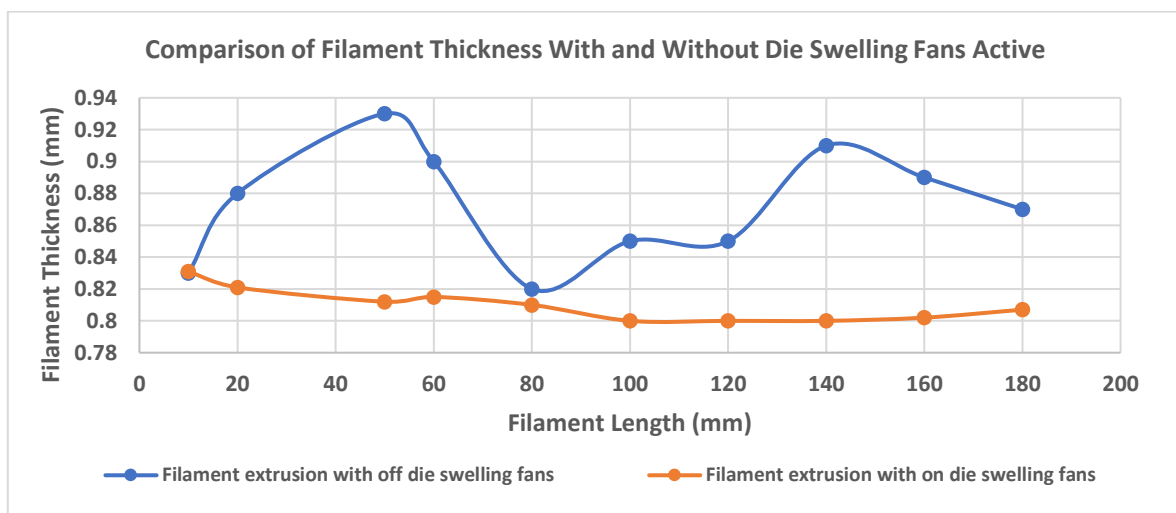


Figure 19 - Filament Diameter Consistency Analysis depicting the comparison of filament thickness measurements over the length of the extruded filament, with and without the activation of die swelling fans. This chart illustrates the impact of the die swelling fans on the uniformity and quality of the extruded filament. The collected data is included in Appendix.

Experiment 2: Three-Dimensional Precision Extrusion

In the domain of additive manufacturing, precision is paramount. The second phase of this investigation delved into the precision extrusion capabilities of the Hybrid Manufacturing machine. A series of structured tests were conducted to examine the machine's proficiency in fabricating a cube of specified dimensions using thermoplastic pellets with a known high viscosity of 10,000 pa.s at 180°C, a challenge for many extrusion systems.

The experiment was meticulously planned, and the slicing parameters were carefully set in Cura to fabricate a cube with layers of 0.2 mm thickness, using a 0.8 mm nozzle for extrusion. The completed cube was then subject to stringent dimensional verification using an electron microscope. Figures 20 and 21, along with Table 6, document the comprehensive measurements taken from the cube during and after the print. The electron microscope images provide a level of detail, crucial for a nuanced understanding of the extruder's precision.

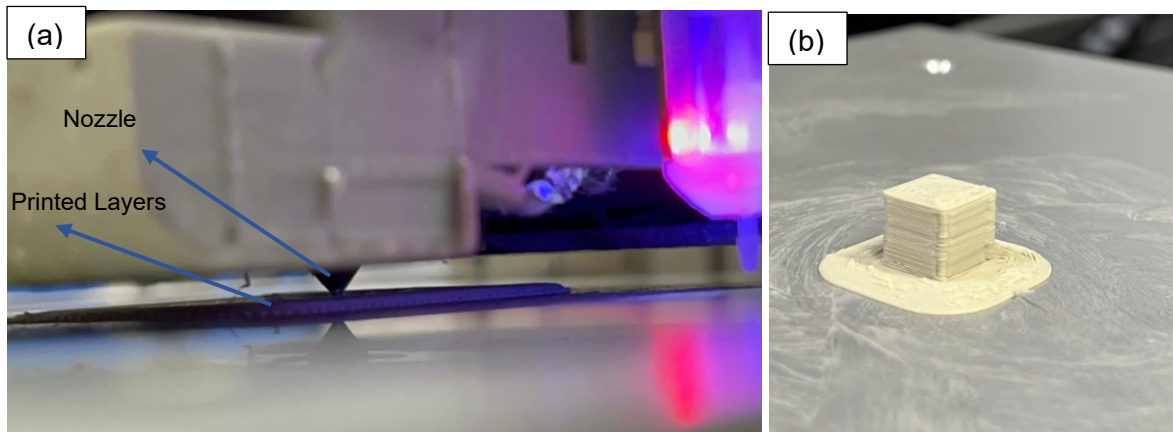


Figure 20 - Precision in Hybrid Manufacturing: (a) Captures the extrusion process as the cube is being printed, illustrating the material deposition, (b) Displays the finished 10x10x10 mm cube post-printing.

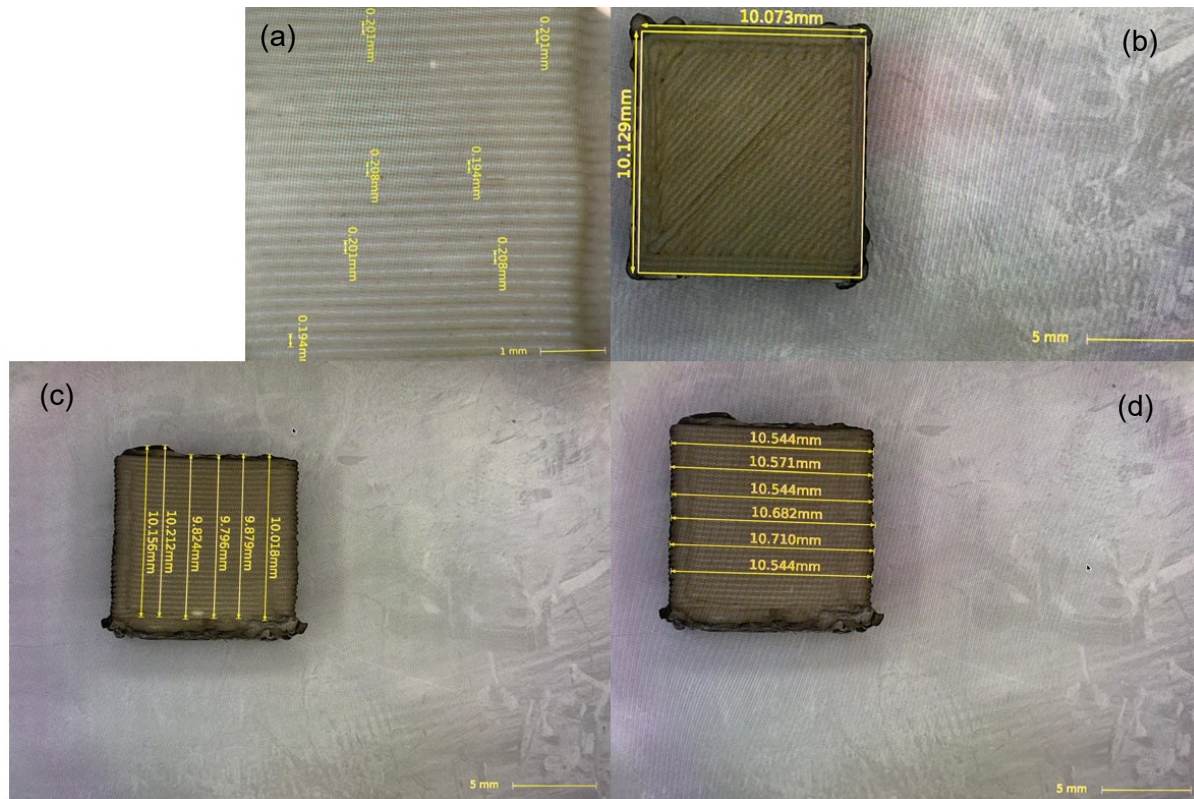


Figure 21 - Electron Microscope Analysis of a 10x10x10 mm Cube: (a) Layer thickness measurements in different printed layers, (b) A top view X-Y measurements, (c) Z-axis height of the cube verification, (d) X-axis side measurement.

Table 6 - Dimensional Accuracy of Extruded Cube: Summary of average deviations in the 10x10x10 mm cube's layer thickness and side measurements in the X, Y, and Z directions, indicating the precision of the extruder in replicating specified dimensions.

	Layer Thickness (mm)	X axis side (mm)	Y axis side (mm)	Z axis side (mm)
Average Measured Value	0.201	10.585	10.5352	9.98
Average Deviation	0.001	0.585	0.5352	0.02

Discrepancies between the measured dimensions and the target cube size of 10 mm in each axis were observed. The layer thickness displayed an average deviation of 0.875-micron, a from the intended layer height of 0.2 mm. The X and Y dimensions of the cube measured 10.585 mm and 10.5352 mm, with deviations of 585 and 535.2 micron respectively, indicating a slightly larger extrusion than expected. The Z dimension came closest to the intended size with a minor 2-micron deviation.

These deviations were analysed, and it was determined that the X-Y positional accuracy, already validated within ± 5 micron, was not the cause. Instead, attention turned to the extruder section. Considering the average error of the extruded filament was found to be 10 microns,

as per previous experiments, it was inferred that the extrusion excess was primarily due to the delay in the retraction system of the extruder. Close examination of the cube's sides (refer to Figures 21 (c) and (d)) revealed this excess, particularly at the start and end points of the extrusion paths. Ignoring the protrusions at the edges, the deviation reduced to approximately 73 and 129 microns on the X and Y axes respectively as shown in Figure 21 (b).

Potential remedies for this issue were suggested, such as redesigning the nozzle shape to square or oval to prevent over-extrusion at the corners. However, a solution would likely involve a combination of hardware modifications and software parameter adjustments, including but not limited to tuning extrusion temperature, motor speed, and die cooling fan. These adjustments fall outside the scope of the current project, which focuses on the proof of concept rather than extensive material development.

The findings from these experiments not only reinforce the robustness of the extruder in handling high-viscosity materials but also, quite significantly, illuminate the strategic merit of this research project. The observed over-extrusion aligns with the well-documented challenges of robocasting, particularly in the context of steps and surface finishes around printed parts. These nuances, while subtle, underline the rationale for the hybrid manufacturing approach developed here, which combines robocasting and CNC machining.

By highlighting these inherent issues in the robocasting process, the experiment underscores the necessity and relevance of integrating precise CNC machining. This integration is not a mere improvement but a fundamental enhancement, addressing the critical need for fine-tuned accuracy and superior surface quality in advanced ceramic components manufacturing. The slight deviations observed serve as a testament to the system's design, which anticipates and accommodates such discrepancies, thus validating the project's foundational hypothesis: hybrid manufacturing is not only viable but essential for advancing the field of additive manufacturing, particularly robocasting.

The compelling results obtained thus far are a promising indication that the pursuit of hybrid manufacturing stands on solid ground. The successful production of the cube is a proof of concept that goes beyond meeting initial expectations. This achievement is a crucial milestone, signalling a pivotal moment where the theoretical aspirations of hybrid manufacturing are realized in tangible, precise, and high-quality outcomes.

4.2.4. Hybrid Manufacturing Synchronization Test

The Hybrid Manufacturing machine's synchronization test effectively demonstrated its ability to manage simultaneous operations across its additive and subtractive manufacturing modules. Initially, a 10x10x10 mm cube was precisely extruded on Bed A, following the specified parameters for temperature and nozzle diameter set in the "3.7.3 Extrusion Capability Experiment 2" section.

Post-extrusion, the automated synchronization system, controlled by the custom-developed software described in section 3.6.6, executed a flawless transition, swapping from the extruder to the CNC machine for immediate machining of the first cube. Concurrently, a second cube

began printing on Bed B, exemplifying the machine's dual-operation capability. This parallel processing not only exemplified the synchronization efficiency but also significantly enhanced the overall manufacturing speed, adhering to the project's efficiency goals. A video demonstrating this process is available for further illustration at [\[link here\]](#).

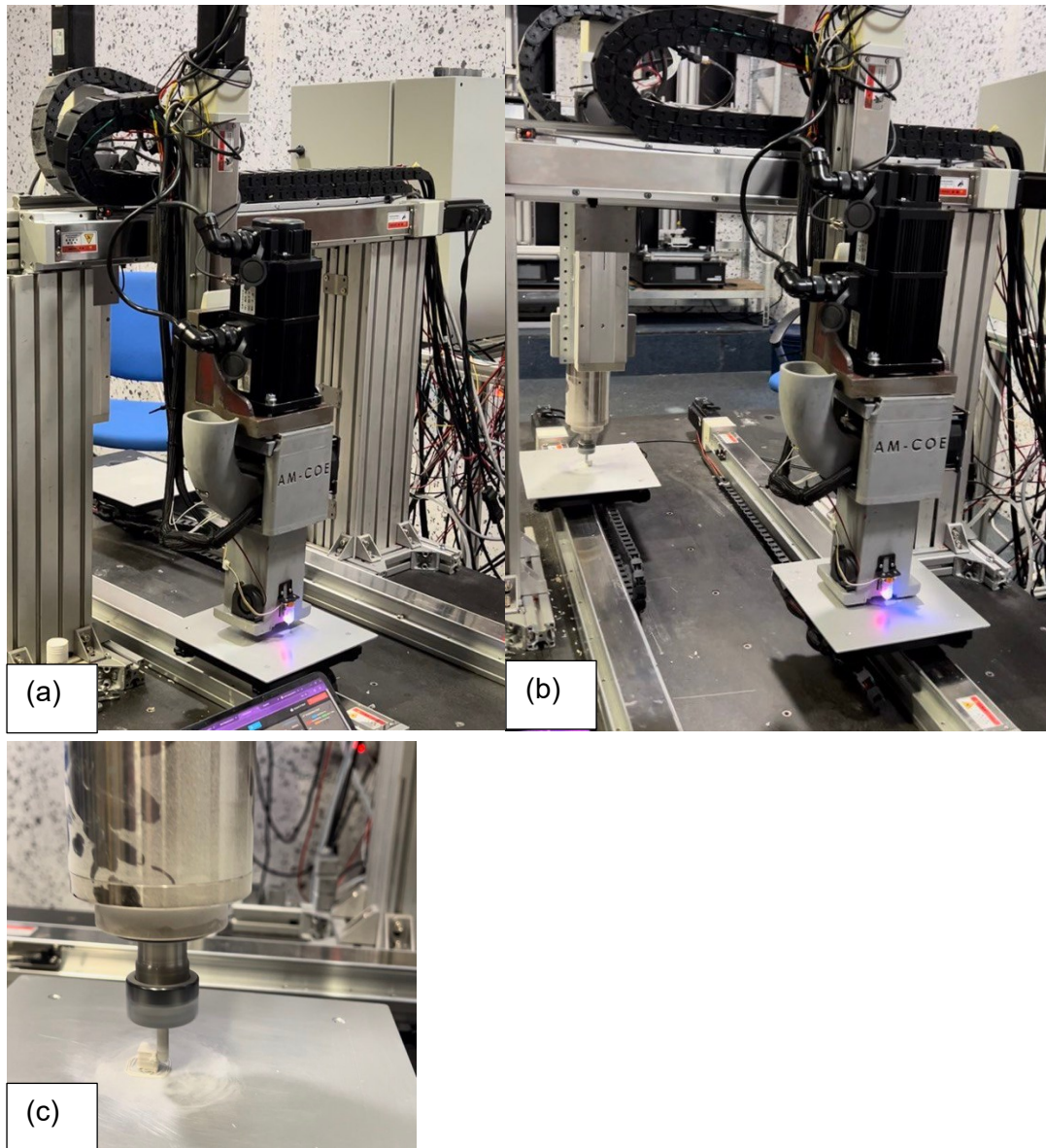


Figure 22 - Dual Operational Mode of the Hybrid Manufacturing Machine: This figure showcases the machine's capabilities in a sequence of operations. Figure (a) displays the initial extrusion printing on Bed A. Figure (b) shows the transition where Bed B begins printing while Bed A undergoes CNC machining, highlighting the machine's seamless operational synchronization. Panel (c) captures simultaneous CNC machining on Bed A alongside printing on Bed B.

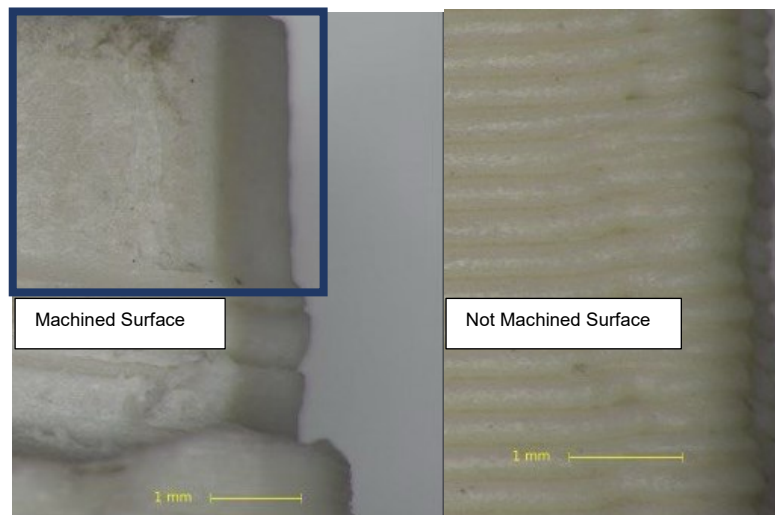


Figure 23- Post-Machining Surface Improvement: This image illustrates the enhanced surface finish of a ceramic component after CNC machining, highlighting the machine's ability to refine the precision and quality of additively manufactured parts.

While this experiment primarily aimed to validate the synchronization and seamless bed swapping, it also opens avenues for further enhancements. Integrating an AI model could optimize these transitions, potentially reducing time delays and enhancing synchronization efficiency.

Importantly, this test demonstrated that the part remained securely attached to the heated bed during machining, due to the controlled bed temperature, which was crucial for maintaining adhesion without compromising the part's integrity during the subtractive process. The CNC machining phase successfully eliminated the steps around the cube, demonstrating the project's effectiveness in overcoming the surface finish and tolerance limitations of the robocasting method of additive manufacturing, as discussed in section 4.2.3's Extrusion Capability Experiment. Figure 23 provides a clear depiction of the improved surface quality post-machining, affirming that the machine's design effectively meets the requirements for manufacturing high-accuracy advanced ceramic components through layer-by-layer hybrid manufacturing. This successful implementation underscores the machine's capability to deliver enhanced surface finishes and precision, confirming the project's objectives and contributing significantly to the field of advanced manufacturing.

5. CONCLUSIONS

This study focused on developing a Hybrid Manufacturing machine that integrates robocasting and CNC machining for high-viscosity ceramic composites, addressing limitations of current methods and enhancing material properties. The machine combines CNC precision with robocasting versatility, evidenced by comprehensive testing. Positional accuracy within ± 5 micron and temperature stability with $\pm 2^\circ\text{C}$ fluctuations were maintained, ensuring consistent material properties. The extruder handled materials up to $40,000 \text{ pa.s}$ at 0.3 s^{-1} , showing a deviation in filament thickness of only 10 micron with die cooling. Dimensional tolerances of the printed cube were noted at 585 micron and 535.2 micron in the X and Y dimensions, respectively, with a 20-micron deviation in the Z dimension. The CNC part aims to perfect surface finishes and reduce tolerances, enhancing additive manufacturing precision. The successful engineering and assembly of the extrusion section, development of XYZ motion control, and establishment of a CNC-extruder synchronization protocol validated the project's aims and objectives. This integration not only demonstrates potential for advanced ceramic production but also contributes significantly to hybrid manufacturing knowledge, confirming the viability of combining additive and subtractive methods for high accuracy and superior surface finishes in ceramics.

6. FUTURE WORK

The development of the hybrid manufacturing machine, which seamlessly integrates robocasting (DIW) with CNC machining, offers a groundbreaking approach to additive manufacturing. This machine, capable of precise layer-by-layer machining and adjustable frequencies, represents a significant advancement in the field, particularly for the production of advanced ceramic components.

Mechanical Design Enhancements: Future iterations of the machine should focus on refining the mechanical design. The development of second and third-generation models could address existing challenges and incorporate enhancements that push the boundaries of what is currently achievable. These improvements are crucial for elevating the machine to commercial standards, making it accessible and beneficial for diverse manufacturing sectors.

Control Systems and Software Development: The control systems of the machine, particularly the PID controllers, can be further optimized to enhance precision and reliability across various operations. Additionally, the existing software requires enhancements to improve its synchronization capabilities. Integrating a machine learning model could significantly advance the synchronization process, making it more intelligent and responsive to the dynamic conditions of hybrid manufacturing.

Material Development: The machine opens new avenues for addressing current limitations in the additive manufacturing of high-viscosity advanced ceramics. There is a vast potential for the development of new ceramic pellet formulations that could lead to the creation of novel material properties and applications. This aspect is particularly exciting as it aligns with the industry's push towards materials that can withstand extreme conditions while maintaining high performance.

Commercialization and Industrial Application: Looking ahead, the ultimate goal is to refine and scale the machine for commercial use. By overcoming current design and operational challenges and enhancing its technological capabilities, the machine can be positioned as a key player in the future of advanced manufacturing. The commercialization of this technology will not only broaden its applicability across industries but also foster innovation in the design and production of complex ceramic components.

In essence, the journey of the developed hybrid manufacturing machine is just beginning. Its innovative approach to combining robocasting with CNC machining lays a solid foundation for future enhancements. These advancements are not only anticipated to refine the technology further but also to usher in a new era of manufacturing precision and efficiency in the ceramics industry and beyond.

7. REFERENCES

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APPENDIX

Detail Drawing of 12-, 16- and 29-mm screw diameters, respectively:

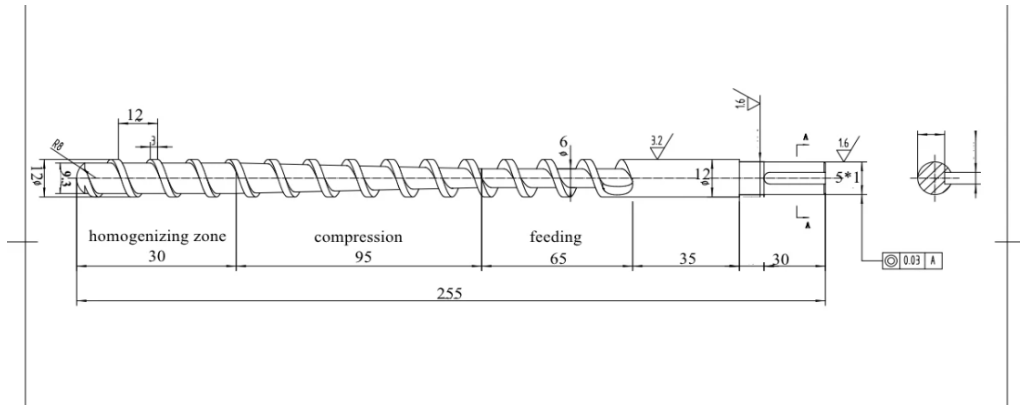


Figure 24. 12mm Single screw detail drawing.

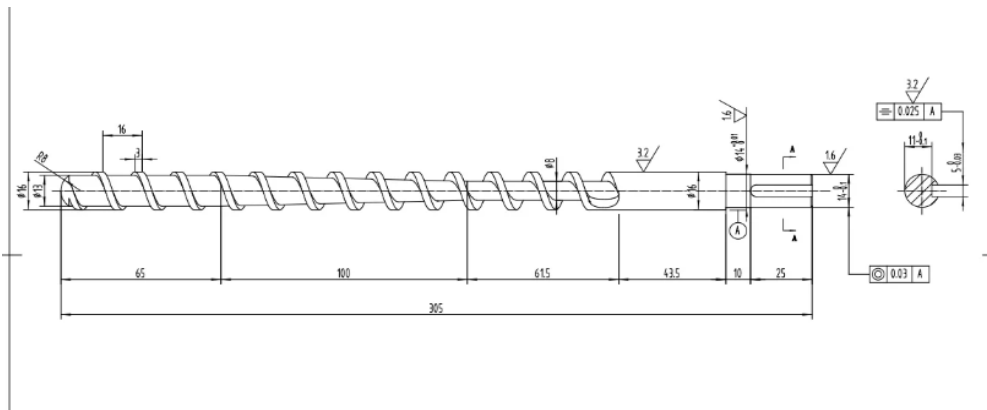


Figure 25. 16mm Single screw detail drawing.

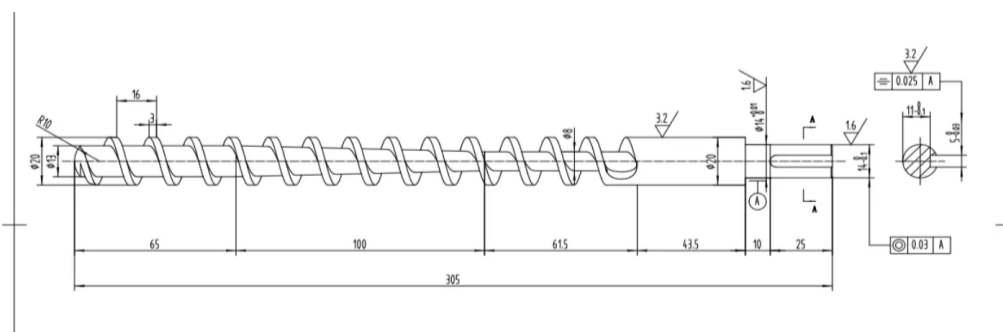


Figure 26. 20mm Single screw detail drawing.

Screw Dimensions:

Table 7. Overview of Screw Dimensions Utilized in Spreadsheet Calculations and Equations

D(mm)	L(mm)	L ₁ (mm)	L ₂ (mm)	L ₃ (mm)	ϵ	L/D	$\emptyset$ (°)	H (mm)
12	255	65	95	30	2.2	21.2	17.65	1.35
16	305	61.5	100	65	2.6	19	17.65	1.5
20	305	61.5	100	65	1.7	15.2	14.28	3.5

'D' represents the screw diameter, 'H' refers to the screw channel depth, 'N' indicates the screw's rotational speed in revolutions per minute (rpm), ' $\emptyset$ ' denotes the flight helix angle in relation to the vertical axis, and ' μ ' signifies the viscosity of the paste being extruded.

Summery Spreadsheet Figure for Screw, Power, and Torque Calculations:

Name	Density	Unit
Ceramic Composite Density	4.50E-06	kg/mm3

12mm Screw Characteristic:

D(mm)	L (mm)	L1 (mm)
12	190	65
L3 (mm)	L3(mm)	Φ (°)
30	30	17.65

Nozzle Characteristic

Nozzle Radius (mm)	Nozzle Channel Length (mm)
0.3	53

Material Charachteristic:

m	n	Viscosity (pa.s)
40000	1	40000

Screw Speed:

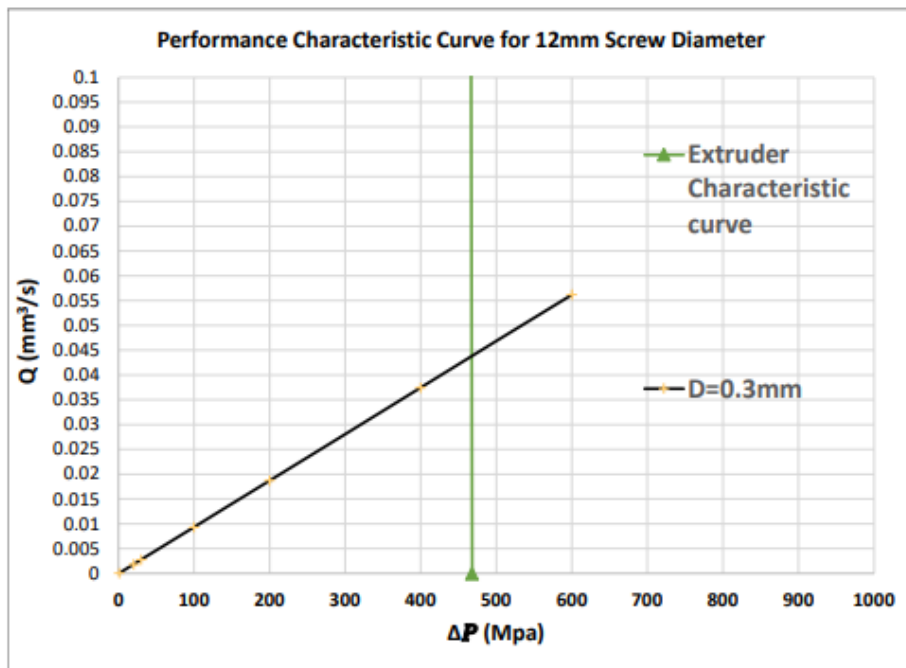
Parameter	Speed (rpm)
N	60

Extrusion Characteristic Line:

Qd (mm3/s)	Pmax (Mpa)
276.8958286	467.8468968

Die Characteristic Line:

Q(ΔP) (mm3/s)	ΔP (Mpa)
9.37279E-05	1
0.001874558	20
0.002811837	30
0.009372789	100
0.018745578	200
0.037491156	400
0.056236733	600



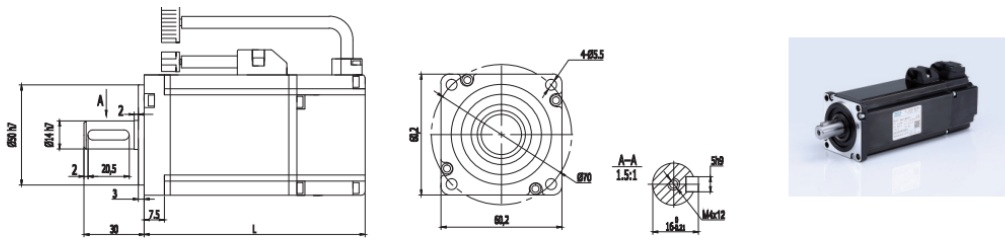
Mechanical Power & Torque Calculation:

Require Power (W)	Require Torque (N.m)
462.3662625	147.2504021

Operating Point Flowrate Analysis & Slowest Extrusion Speed Analysis:

Q (mm³/s)	Extrusion Velocity (mm/s)
0.045	0.63

X-Y-Z AC Servo Motor Specification:



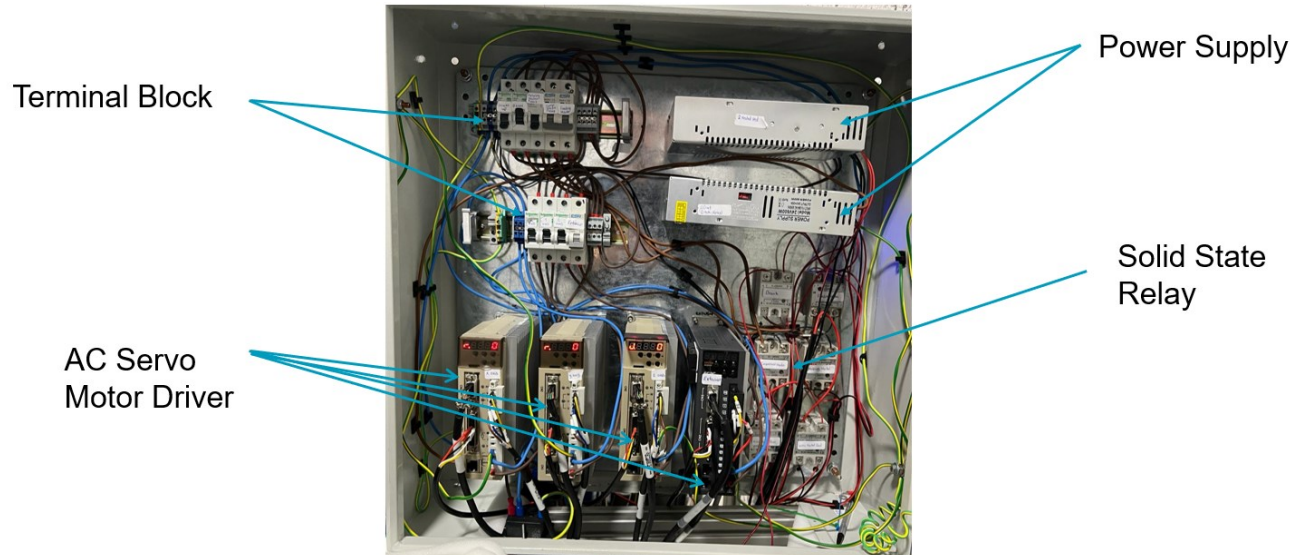
⦿ 技术参数 / SPECIFICATIONS

电机型号 MOTOR MODEL		60AST-M00630					60AST-M01330					60AST-M01930														
标准命名		MN060-P63FFDI2-01B000					MN060-013FFDI2-01B000					MN060-019FFDI2-01B000														
额定功率 RATED POWER(W)							400																			
额定线电压 RATED VOLTAGE(V)							220																			
额定线电流 RATED CURRENT(A)							2.8																			
额定转速 RATED SPEED(RPM)							3000																			
额定力矩 RATED TORQUE(N.m)							1.27																			
峰值力矩 PEAK TORQUE(N.m)							3.9																			
反电势 VOLTAGE CONSTANT(V/1000r/min)							29.6																			
力矩系数 TORQUE COEFFICIENT(N.m/A)							0.45																			
转子惯量 ROTOR INERTIA(KG.M²)							0.29x10 ⁻⁴																			
绕组（线间）电阻 LINE-LINE RESISTANCE（Ω）							2.35																			
绕组（线间）电感 LINE-LINE INDUCTANCE（mH）							14.5																			
电气时间常数 ELECTRICAL TIME-CONSTANT（ms）							6.17																			
重量 WEIGHT（KG）							1.6																			
编码器线数 ENCODER LINE NUMBER(PPR)							2500																			
电机绝缘等级 INSULATION CLASS							ClassF																			
防护等级 SAFETY CLASS							IP65																			
使用环境 ENVIRONMENT							：-20℃ ~ +40℃ 环境温度；相对湿度<90%（不结 RATURE:-20℃~+40℃ HUMIDITY BELOW 90%F																			
		增量编码器电机绕组引线					U(红)					V(黄)					W(蓝)					PE(黄绿)				
电机绕组插座 MOTOR WINDING PLUE		绝对值编码器电机绕组引线					PE(黄绿)					U(红)					V(蓝)					W(黄)				
		插座编号					1					2					3					4				
增量编码器插座（15芯） INCREMENTAL ENCODER SOCKET(15PIN)		信号引线					5V 0V B Z- U Z U- A V W V- A- B- W-										PE									
		插座编号					2 3 4 5 6 7 8 9 10 11 12 13 14 15										1									
绝对值编码器插座（7芯） ABSOLUTE ENCODER SOCKET(7PIN)		信号引线					E- E+ SD- 0V SD+ +5V										PE									
		插座编号					2 3 4 5 6 7										1									
L不带抱闸		116										141										169				
L带永磁抱闸		164										189										217				
L带电磁抱闸		154										179										207				

[illegible]

Motor Model	EC-F4CC120V-M2XM+A
Rated power(kw)	1.2
Rated voltage(V)	220
Rated current(A)	5.0
Rated Speed(RPM)	3000
Rated Torque(N.m)	4
Peak Torque(N.m)	12
Weight(g)	7235

Electrical Setup for the Hybrid Manufacturing Machine: This image depicts the electrical box responsible for powering the controllers and supporting the additive manufacturing component of the Hybrid Manufacturing machine. It includes detailed wiring configurations, ensuring efficient power distribution and operational stability for the machine's additive section.



AC Servo Motor Driver:



XP200-20PL (1174g)	
Input Voltage	220VAC-240VAC
Input Phase	1Phase or 3Phase
Input Frequency	50/60Hz
Output max current	5.5A

Experiment 1 data in section 4.2.3 with off die fans:

No. Area	Filament Diameter (mm)	Filament Length (mm)
1	0.83	10
2	0.88	20
3	0.93	50
4	0.9	60
5	0.82	80
6	0.85	100
7	0.85	120
8	0.91	140
9	0.89	160
10	0.87	180
Average	0.873	

Experiment 1 data in section 4.2.3 with on die fans:

No. Area	Filament Diameter (mm)	Filament Length (mm)
1	0.831	10
2	0.821	20
3	0.812	50
4	0.815	60
5	0.81	80
6	0.8	100
7	0.8	120
8	0.8	140
9	0.802	160
10	0.807	180
Average	0.8098	

Experiment 2 data in section 4.2.3:

No. measurement	layer thickness (mm)	X axis side measurement (mm)	Y axis side measurement (mm)	Z axis side measurement (mm)
1	0.201	10.544	10.535	10.156
2	0.201	10.571	10.61	10.212
3	0.194	10.544	10.546	9.824
4	0.208	10.682	10.5	9.796
5	0.208	10.71	10.43	9.878
6	0.201	10.544	10.583	10.018
7	0.194	10.5	10.543	
8	0.2			
Average	0.200875	10.585	10.53528571	9.980666667
Average Deviation	0.00875	0.585	0.535	0.02

Configuration G-Code for the Additive Manufacturing Mainboard:

NOTE: This G-Code is not the firmware source code itself; instead, it is used to configure the firmware settings specifically for this machine. The customised source code is too long to be put here.

```

46 ; Extruders
47 M584 E0.4 ; set extruder mapping
48 M350 E128 I0 ; configure microstepping without interpolation
49 M92 E800; configure steps per mm
50 M566 E120 ; set maximum instantaneous speed changes (mm/min)
51 M203 E30000 ; set maximum speeds (mm/min)
52 M201 E1250 ; set accelerations (mm/s^2)
53
54 ; Kinematics
55 M669 K0 ; configure Cartesian kinematics
56
57 ; Probes
58 M950 S0 C"io7.out" ; create servo #0 for BLtouch
59 M558 K0 P9 C"io7.in" H5 F120 T6000 ; configure BLTouch probe via slot #0
60 G31 X2 Y0 Z0 P25 ; set Z probe trigger value, offset and trigger height
61 M574 Z1 S2 ; configure Z axis endstop
62
63
64
65
66 ; Endstops
67 M574 X2 P"io1.in" S1 ; configure X axis endstop
68 M574 Y2 P"io2.in" S1 ; configure Y axis endstop
69
70
71 ; Sensors
72 M308 S0 P"temp0" Y"thermistor" A"Heated Bed cnc" T100000 B4725 C7.06e-8 ; configure sensor #0 temp 0, 1 roye board vase thermistor
73 M308 S1 P"temp1" Y"thermistor" A"Heated Bed robo" T100000 B4725 C7.06e-8 ; configure sensor #1
74 ; Sensors
75 M308 S2 P"temp2" Y"thermistor" A"Zone 1 Temp" T100000 B4725 C7.06e-8 ; configure sensor #0 temp 0, 1 roye board vase thermistor
76 M308 S3 P"temp3" Y"thermistor" A"Zone 2 & 3 Temp" T100000 B4725 C7.06e-8 configure sensor #0 temp 0, 1 roye board vase thermistor
77
84 ;M308 S4 P"spi.cs3" Y"thermocouple-max31856" A"Metering Temp" ; configure sensor #4
85
86
87
88
89 ; Heaters
90 ; Heated bed
91 M950 H0 C"out0" T0 ; create heater #0
92 M143 H0 P0 T0 C0 S110 A0 ; configure heater monitor #0 for heater #0
93 ;M307 H0 R1.0 D5.5 E1.35 K0.56 B0 ; configure model of heater #0
94 ; Adjust Heater 1 model for slower temperature rise
95 M307 H0 R1.0 D5.5 E1.35 K0.56 B0 ; You can further increase D if needed
96
97 ; Relax the heater fault conditions even more
98 M570 H0 P180 ; Set the heater fault detection period to 24 hours (86400 seconds)
99
100
101
102
103 M950 H1 C"out1" T1 ; create heater #1
104 M143 H1 P0 T1 C0 S110 A0 ; configure heater monitor #0 for heater #1
105 ;M307 H1 R1.0 D5.5 E1.35 K0.56 B0 ; configure model of heater #1
106 ; Adjust Heater 1 model for slower temperature rise
107 M307 H1 R1.0 D5.5 E1.35 K0.56 B0; You can further increase D if needed
108
109 ; Relax the heater fault conditions even more
110 M570 H1 P180 ; Set the heater fault detection period to 24 hours (86400 seconds)
111
112
113
114

```

```
12 M552 P192.168.8.109 S1 ; configure Ethernet adapter
13 M586 P0 S1 ; configure HTTP
14
15 ; Motor Idle Current Reduction
16 M906 I30 ; set motor current idle factor
17 M84 S30 ; set motor current idle timeout
18 ;bed swap
19
20 ; Define IO_8 as a general purpose output
21 M950 P8 C"io8.out.iso" ; Set pin 8 as an output, '!' indicates active low if needed
22 M42 P8 S0
23 ; Define IO_5 as a general purpose output
24 M950 P5 C"io5.out.iso" ; Set pin 5 as an output, '!' indicates active low if needed
25 M42 P5 S0
26
27
28 ; External Drivers
29 M569 P0.3 S0 R0 T0.2:0.2:0.1:0 ; driver 0.0 goes forwards and requires an active-low enable signal (X axis)
30 M569 P0.1 S1 R0 T0.2:0.2:0.1:0 ; driver 0.1 goes forwards and requires an active-low enable signal (Y axis)
31 M569 P0.2 S0 R0 T0.2:0.2:0.1:0 ; driver 0.2 goes forwards and requires an active-low enable signal (Z axis)
32 M569 P0.4 S0 R0 T0.2:0.2:10:0
33 ;M569 P0.4 S1 R0 T0.2:0.2:10:0 ; driver 0.4 goes forwards and requires an active-low enable signal (extruder 0)
34
35 ; Axes
36 M584 X0.3 Y0.1 Z0.2; set axis mapping
37 M350 X128 Y128 Z128 I0 ; configure microstepping without interpolation
38 M92 X2000 Y2000 Z2000 ; configure steps per mm
39 M208 X0:900 Y0:600 Z0:300 ; set minimum and maximum axis limits
40 M566 X100000 Y10000 Z5 ; set maximum instantaneous speed changes (mm/min)
41 M203 X10000 Y10000 Z30000 ; set maximum speeds (mm/min)
42 M201 X40 Y20 Z3 ; set accelerations (mm/s^2)
```



```
118
119 M950 H2 C"out3" T2 ; create heater #2 feeding
120 M143 H2 P0 T2 C0 S400 A0 ; configure heater monitor #0 for heater #2
121 M307 H2 R1.0 D5.5 E1.0 K0.56 B0 ; configure model of heater #2
122 M570 H2 P180 ; Set the heater fault detection period to 24 hours (86400 seconds)
123
124
125 ; Heaters
126 ; Heater 3 - Using Sensor S3
127 M950 H3 C"out4" T3
128 M143 H3 S400
129 M307 H3 R1.0 D5.5 E1.35 K0.56 B0
130 M570 H3 P180 ;SINA
131
132
133 ; Heater 4 - Using Sensor S3
134 M950 H4 C"out5" T3
135 M143 H4 S400
136 M307 H4 R1.0 D5.5 E1.35 K0.56 B0
137 M570 H4 P180
138
139
140 ; Heated beds dadn har heater be bed p index h heater
141 M140 P0 H0 S20; configure heated bed #0
142 M140 P1 H1 S20; configure heated bed #1
143
144
145 ; Fans
146 M950 F1 C"out6" ; create fan #1
147 M106 P1 C"Die Fan 1" S0 ; fan #1 is now manual control, initially off
148
149 M950 F2 C"out7" ; create fan #2
150 M106 P2 C"Die Fan 2" S0 ; fan #2 is now manual control, initially off
```

```
152 M950 F3 C"out8" ; create fan #3
153 M106 P3 C"Heat Sink Fan" S0 ; fan #3 is now manual control, initially off
154
155
156
157
158 ; Miscellaneous
159 M501 ; load saved parameters from non-volatile memory
160
161 ; Tools
162 ; Tool 0 - using heaters H2, H3, H4
163
164
165 ;assign pin for break for z
166 M950 P0 C"out2" ; Assign GPIO port 0 to out2 for brake control
167 M42 P0 S1 ; Set GPIO port 0 (out2) to high (on) to engage the brake
168
169 ;assign pin for break for extrusion
170 M950 P1 C"io6.out.iso" ; P0 is just a placeholder for the output number, "!" before the pin name inverts the signal if needed
171 M42 P1 S1 ; Set port P0 high to disengage brake
172
173
174 ; Tools
175 M563 P0 D0 H2:3:4 F3 ; define tool 0
176 G10 P0 X0 Y0 Z0 ; set tool 0 axis offsets
177 G10 P0 R0 S0
178 M302 ; Report current state
179 M302 P1 ; Allow cold extrusion
180 M375 ; Load height map ; set initial tool 0 active and standby temperatures to 0C
181 T0
182
```